Type MS · Lengths in a 3D solid using right-angled triangles

Answer 1 of 22 · Question 11 part a · 3 marks

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11(a)	Let the intersection of AC and DB be X $[AC =]\sqrt{6^2+6^2}\left[=\sqrt{72} \Rightarrow AX = 3\sqrt{2}\right]$ oe or $[AX =]\sqrt{3^2+3^2}\left[=\sqrt{18}=3\sqrt{2}\right]$ $[h =]\sqrt{12^2-"\left(3\sqrt{2}\right)"^2} = 3\sqrt{14}$ oe*	M1 M1A1 cso [3]
ALT1	Let Z be the midpoint of AB $[EZ =]\sqrt{12^2-3^2}\left[=\sqrt{135}=3\sqrt{15}\right]$ oe $[h =]\sqrt{("3\sqrt{15}")^2-(3)^2} = 3\sqrt{14}$ oe*	M1 M1A1 cso [3]
ALT2	$[h =]\sqrt{3^2+3^2+h^2} = 12$ or $3^2+3^2+h^2 = 144$	M1M1 A1
(b)	Required angle $= [\angle EAX]\sin^{-1}\left(\frac{3\sqrt{14}}{12}\right)$ or $\tan^{-1}\left(\frac{3\sqrt{14}}{3\sqrt{2}}\right)$ or $\cos^{-1}\left(\frac{3\sqrt{2}}{12}\right) = 69.295...^\circ$ or $\cos^{-1}\left(\frac{\left(6\sqrt{2}\right)^2+12^2-12^2}{2\times 6\sqrt{2}\times 12}\right) \approx 69.3^\circ$	M1A1 [2]

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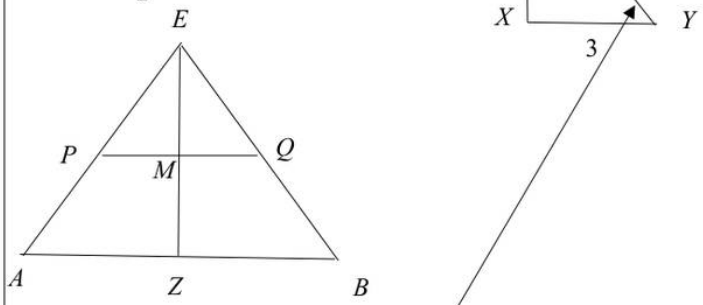
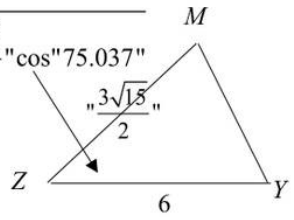
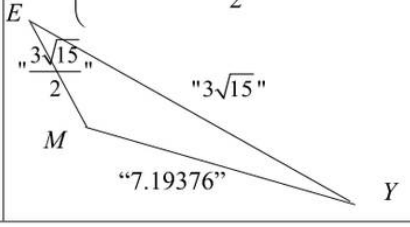
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Part	Mark	Notes
(a)	M1	For correctly applying Pythagoras theorem to find AC or AX Full method, must include square rooting.
	M1	For correctly applying Pythagoras theorem to find h Allow follow through of their AX/AC. Full method, must include square rooting. Not a dependent method mark, but must see M1 or be correctly using $3\sqrt{2}$ M0 M1 A0 will occur. We will permit eg $3\sqrt{2}^2$ if the intention is clear
	A1*cso	For the correct length only. Minimum steps shown, no errors or omissions.
ALT1	M1	For correct applying Pythagoras theorem to find EZ Full method, must include square rooting.
	M1	For correctly applying Pythagoras theorem to find h Allow follow through of their EZ. Full method, must include square rooting. Not a dependent method mark, but must see M1 or be correctly using $3\sqrt{15}$ M0 M1 A0 will occur. We will permit eg $3\sqrt{15}^2$ if the intention is clear
	A1*cso	For the correct length only. Minimum steps shown, no errors or omissions.
ALT2	M1M1	For a fully correct statement using 3D Pythagoras, each side shown. M0 M0 for any part wrong
	A1*cso	For the correct length only. Minimum steps shown, no errors or omissions.
(b)	M1	For using the given value of h and applying correct trigonometry to find the required angle. Allow ft of $3\sqrt{2}$ if from correct method. If any other method used, the **method** must be complete and correct ie leaving as sin = is not complete. Note equivalent angles (because of symmetry) eg $\angle EBD$
	A1	For the correct angle awrt 69.3°

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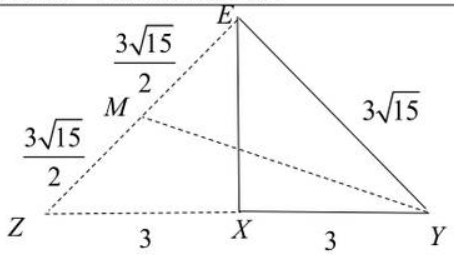
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(c)	Let M be the midpoint of PQ, Y be the midpoint of DC and Z be the midpoint of AB	
	$[EY = EZ =]\sqrt{3^2 + \left(3\sqrt{14}\right)^2}\ [= 3\sqrt{15}]$	M1
	$[EM =]\frac{"3\sqrt{15}"}{2}$	M1
		
	Angle between planes EAB & ABC =	
	$\left[[\angle EYX =]\tan^{-1}\left(\frac{3\sqrt{14}}{3}\right)\left[= 75.0367^\circ\right]\right.$ $\left.\text{or } \cos^{-1}\left(\frac{"\left(3\sqrt{15}\right)"^2 + 6^2 - "\left(3\sqrt{15}\right)"^2}{2\times"3\sqrt{15}"\times 6}\right) \text{ or } \cos^{-1}\left(\frac{3}{"3\sqrt{15}"}\right) \text{ or } \sin^{-1}\left(\frac{3\sqrt{14}}{"3\sqrt{15}"}\right)\right]$	M1
	$[MY =]\sqrt{6^2 + \left("\frac{3\sqrt{15}}{2}"\right)^2 - 2\times 6\times "\frac{3\sqrt{15}}{2}"\cos"75.037"}$ $= 7.19376...$	A1 (M1 on ePen)
		
	[Required angle = $\angle EMY$ =]	
	$\cos^{-1}\left(\frac{\left("\frac{3\sqrt{15}}{2}"\right)^2 + "7.19376"^2 - \left("3\sqrt{15}"\right)^2}{2\times"\frac{3\sqrt{15}}{2}"\times"7.19376"}\right) = 126.3133 = 126$	dddM1 A1
		
		[6]

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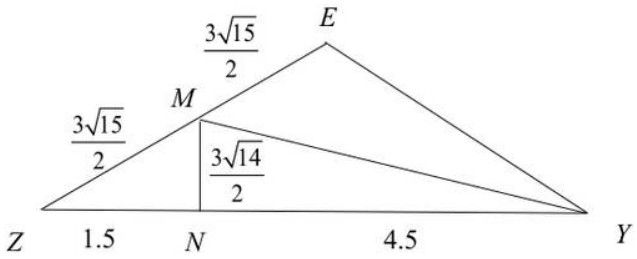
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3rd M mark and 1st A mark – ALTs to find MY		
	 $\left[\left[\angle ZEY = 2\times\angle XEY =\right] 2\times\tan^{-1}\left(\frac{3}{3\sqrt{15}}\right)\left[= 2\times 14.9632^{\circ}\right]\right]$ or $\left[\left[\angle ZEY =\right]\cos^{-1}\frac{"\left(3\sqrt{15}\right)"^2+"\left(3\sqrt{15}\right)"^2-"(6)"^2}{2\times"3\sqrt{15}"\times"\left(3\sqrt{15}\right)"}\left[= 29.9264\right]\right]$	M1
	$MY = \sqrt{\left("3\sqrt{15}"\right)^2+\left("\frac{3\sqrt{15}}{2}"\right)^2-2\times"3\sqrt{15}"\times"\frac{3\sqrt{15}}{2}"\cos 29.9264}$ $= 7.19376...$	A1
		Total 11 marks

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Question	Scheme	Marks
11c ALT	Dropping a perpendicular from M to the base of the pyramid, which hits at point N  $(MN=)\frac{3\sqrt{14}}{2}$ $\left(\text{Note: } MZ=\sqrt{\left(\frac{3\sqrt{14}}{2}\right)+1.5^2}=\frac{3\sqrt{15}}{2}\right)$	M1
	$(\angle ZMN=)\tan^{-1}\left(\frac{1.5}{\frac{3\sqrt{14}}{2}}\right)$ or $\cos^{-1}\left(\frac{\frac{3\sqrt{14}}{2}}{\frac{3\sqrt{15}}{2}}\right)$	M1
	or $\sin^{-1}\left(\frac{1.5}{\frac{3\sqrt{15}}{2}}\right)(=14.96321743)$	
	$(\angle YMN=)\tan^{-1}\left(\frac{4.5}{\frac{3\sqrt{14}}{2}}\right)(=38.7220711)$	M1
	$(\angle YMZ=14.96321743+38.7220711=)53.68528853$	A1
	$180-"53.68528853"=126.3$	dddM1A1 [6]

Part	Mark	Notes
ALT	M1	For correctly stating (explicitly or in working) the height of MN
	M1	For the full and correct **method** to find $\angle ZMN$ They may follow through MZ if found and from a correct method.
	M1	For the full and correct **method** to find $\angle YMN$ They may follow through MZ if found and from a correct method.
	A1	For the correct angle $\angle YMZ$ allow awrt 39
	dddM1	Dependent on all previous method marks, for subtracting their angle from 180.
	A1	For awrt 126°

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Answer 2 of 22 · Question 3 part a · 3 marks

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3(a)	$AM = \sqrt{6^2+8^2} = 10$	M1
	$AE = \sqrt{14^2+10^2} = \sqrt{296} = 17.20... = 17.2\,\text{cm}$	M1A1 (3)
(b)	$\tan\phi = \frac{EM}{MA} = \frac{14}{10}$, $\phi = 54.46... = 54.5°$ or using another trig function	M1A1ft,A1(3)
(c)	$\tan\theta = \frac{EM}{\frac{1}{2}CD} = \frac{14}{8}$, $\theta = 60.255...° = 60.3°$	M1A1ft,A1 (3) [9]

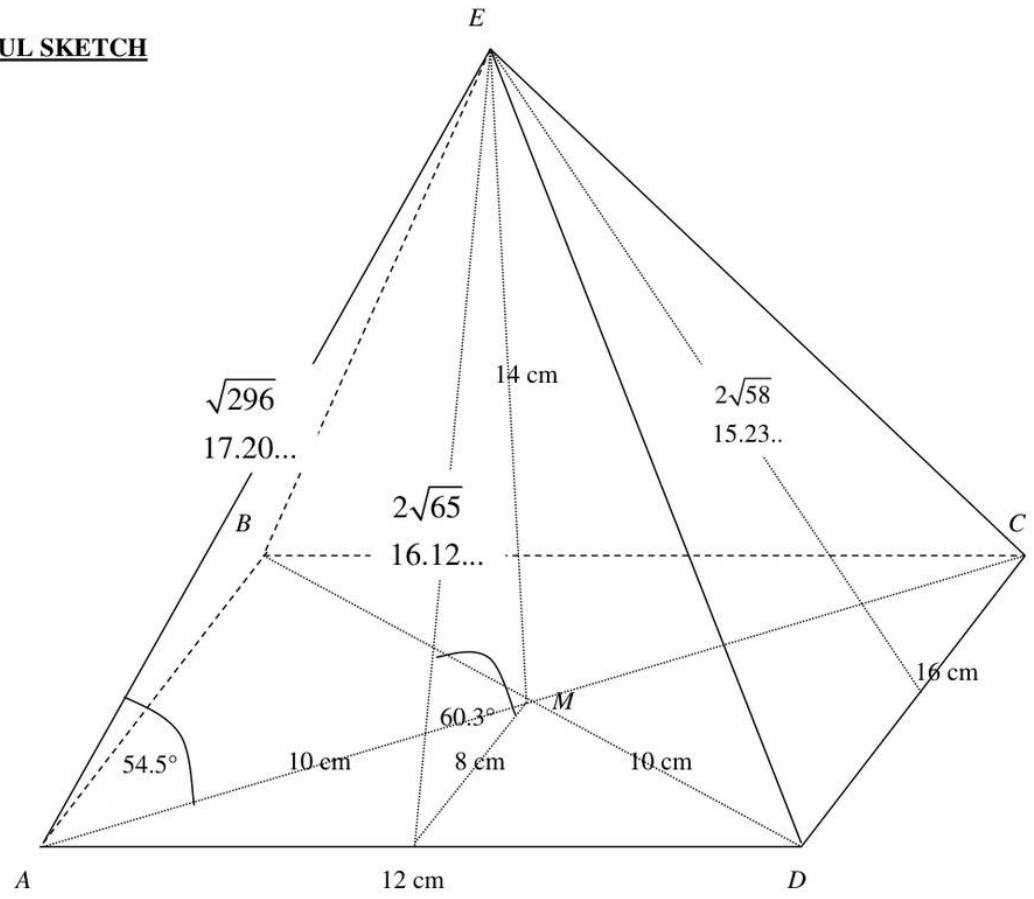
Part	Mark	Notes
(a)	M1	Applies Pythagoras theorem to find the length of AM $AM = \sqrt{6^2+8^2} = 10$ or $AM = \frac{\sqrt{12^2+16^2}}{2} = 10$
	M1	Applies Pythagoras to find the length of one of the sloping edges $AE = \sqrt{14^2+10^2} = \sqrt{296} = ...$
	A1	For the correct length of either AE, DE, CE or BE $AE = 17.2\,\text{cm}$ rounded correctly
	ALT	
	M1M1	Applies Pythagoras in 3D $AE = \sqrt{14^2+6^2+8^2} = \sqrt{296} = ...$
(b)	M1	For applying any acceptable trigonometry to find the required angle. $\tan\phi = \frac{EM}{MA} = \frac{14}{10}$, or $\sin\phi = \frac{14}{\sqrt{296}}$, or $\cos\phi = \frac{10}{\sqrt{296}} \Rightarrow \phi = ...$
	A1ft	For the correct trigonometry if they use sine or cosine following through their $\sqrt{296}$
	A1	Required angle = 54.5° Rounded correctly
(c)	M1	For applying trigonometry to find the required angle. $\tan\theta = \frac{EM}{\frac{1}{2}CD} = \frac{14}{8} \Rightarrow \theta = ...$ OR The length of the perpendicular from E to the mid-point of AD is $\sqrt{260}$ $\sin\theta = \left(\frac{14}{'\sqrt{260}'}\right)$, or $\cos\left(\frac{8}{'\sqrt{260}'}\right) \Rightarrow \theta = ...$
	A1ft	Ft their $\sqrt{260}$
	A1	$\theta = 60.3°$
Rounding: Penalise rounding only the first time it occurs in either (b) or (c)		

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USEFUL SKETCH



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Question Number	Scheme	Marks

Type MS · Lengths in a 3D solid using right-angled triangles

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Part	Mark	Notes
(a)	M1	Divides through $x^3-4x^2+5=0$ by x^2 and rearranges to achieve as a minimum $x+\frac{5}{x^2}=k$ where k is a constant $\left[x-4+\frac{5}{x^2}=0\Rightarrow x+\frac{5}{x^2}=4\right]$
	M1	**Draw the line $y=k$** following through their value for k No line is M0
	A1	For the two values of $x=1.4$ and $x=3.6$ Condone answers given as coordinates provided they are completely correct. (1.4, 4) and (3.6, 4) Require both M marks for this mark.
(b)	M1	For setting $x+\frac{5}{x^2}=Ax+B\Rightarrow x^3+5=Ax^3+Bx^2\Rightarrow Ax^3-x^3+Bx^2-5=0$, and equating coefficients with x^3-x^2-5 $x^3(A-1)+Bx^2-5\equiv x^3-x^2-5$ to achieve as a minimum $A=(\pm 2)$, $B=(\pm 1)$
	A1	For the correct straight line $y=2x-1$
	ALT – to find the line $y=2x-1$	
	M1	Divides through $x^3-x^2-5=0$ by x^2 and rearranges the equation to achieve as a minimum $\Rightarrow \pm 2x\pm 1=x+\frac{5}{x^2}$
	A1	For the correct straight line $y=2x-1$
	dM1	**Draws their $y=2x-1$ on the graph** and locates the point of intersection. Please check that they draw their line correctly. Coordinates for you to check are (0.5, 0) and (2.5, 4) No line is M0 This mark is dependent on the first M mark in (b)
	A1	For the correct value of $x=2.1$ [allow $x=2.2$] Can only score this mark from M1A1M1 Do not accept the answer given as coordinates.

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Question Number	Scheme	Marks
5(a)	Gradient $PR = \frac{6}{-6} = -1$, Gradient $QS = \frac{8}{8} = 1$	M1A1
	Product $= -1 \Rightarrow$ perpendicular	A1 (3)
(b)	(i) $PR = \sqrt{6^2 + 6^2} = 6\sqrt{2}$ $\left(= \sqrt{72}\right)$	M1
	(ii) $QS = \sqrt{8^2 + 8^2} = 8\sqrt{2}$ $\left(= \sqrt{128}\right)$	A1 (2)
(c)	Area $= \frac{1}{2}$ "$6\sqrt{2}$"×"$8\sqrt{2}$" $= 48$ $\left(\text{units}^2\right)$	M1A1 (2)
		[7]

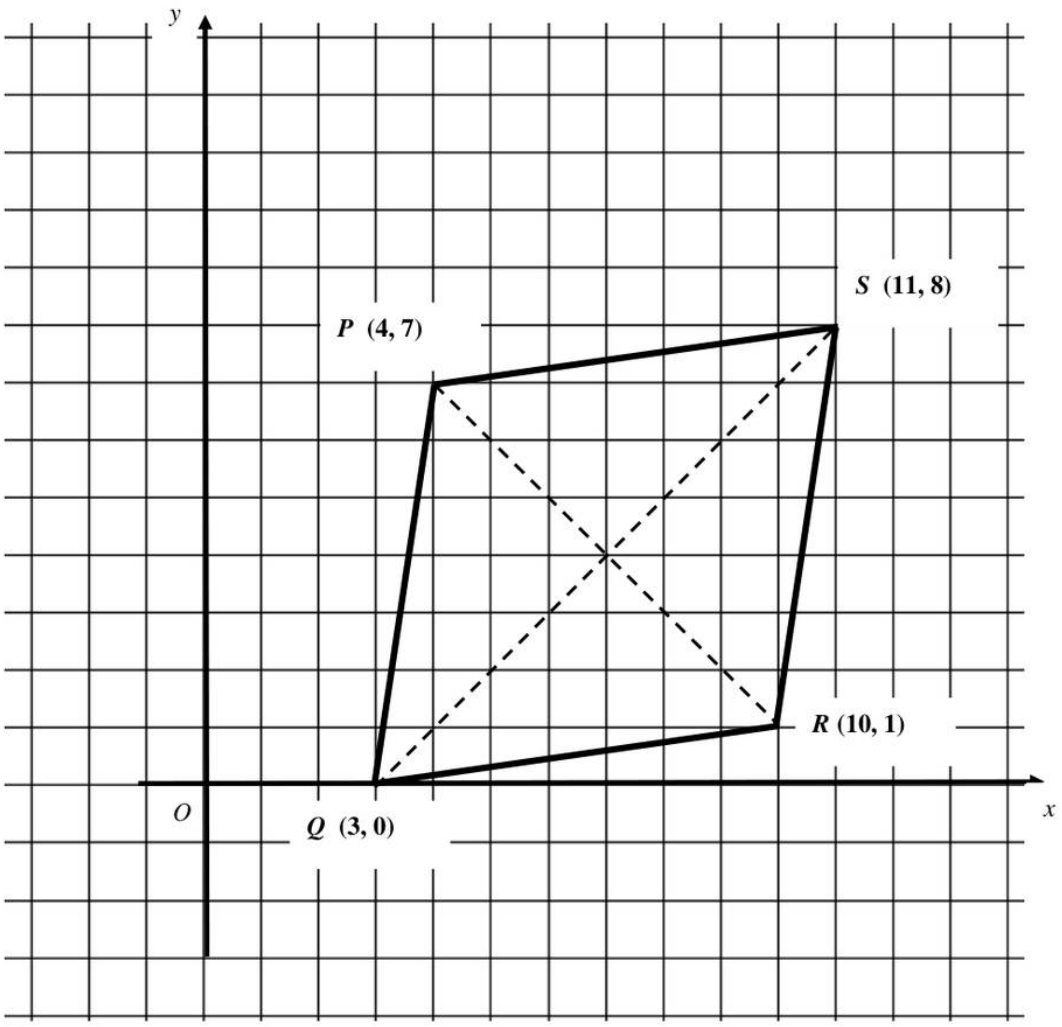
Part	Mark	Notes
(a)	M1	Finds the gradient of PR and QS using a correct method. This may be on a diagram. Gradient $PR = \frac{7-1}{4-10} = \frac{6}{-6} = -1$, Gradient $QS = \frac{8-0}{11-3} = \frac{8}{8} = 1$
	A1	Both gradients correct Gradient $PR = -1$, Gradient $QS = 1$
	A1	Finds the product of the two gradients with a statements that as the product = –1 then the lines are perpendicular.
(b)	M1	For **either** $PR = \sqrt{6^2 + 6^2} = \sqrt{72}$ or $6\sqrt{2}$ **OR** $QS = \sqrt{8^2 + 8^2} = \sqrt{218}$ or $8\sqrt{2}$ correct
	A1	For **both** $PR = \sqrt{6^2 + 6^2} = \sqrt{72}$ or $6\sqrt{2}$ **AND** $QS = \sqrt{8^2 + 8^2} = \sqrt{218}$ or $8\sqrt{2}$ correct
(c)	M1	$PQRS$ is a kite so $= \frac{1}{2}$ "$6\sqrt{2}$"×"$8\sqrt{2}$" $= \ldots$
	A1	Area = 48 $\left(\text{units}^2\right)$
	ALT Uses determinants	
	M1	Area $= \frac{1}{2}\begin{pmatrix} 4 & 3 & 10 & 11 & 4 \\ 7 & 0 & 1 & 8 & 7 \end{pmatrix}$ $= \frac{1}{2}\left(\left[4\times0+3\times1+10\times8+11\times7\right]-\left[3\times7+10\times0+11\times1+4\times8\right]\right)$ $= \ldots$ Allow one slip in a product.
	A1	Area = 48 $\left(\text{units}^2\right)$

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Question Number	Scheme	Marks
6(a)	$a = S_1 = 1(15+2\times1) = 17$	B1
	$S_2 = 2(15+2\times2)(=38) = 2a+d$	M1A1
	$2\times17+d=38 \Rightarrow d=4$	A1 (4)
(b)	20th term $= a+19d = 17+19\times4 = 93$	M1A1 (2)
(c)	$S_{2p}-2S_p = 1+S_{p-1}$	
	$2p(15+4p)-2p(15+2p) = 1+(p-1)(13+2p)$	M1
	$2p^2-11p+12=0$	A1
	$(2p-3)(p-4)=0 \Rightarrow p=4$ $\left(p\neq\frac{3}{2};\text{ may not be seen}\right)$	M1A1 (4)
		[10]

Part	Mark	Notes
(a)	B1	For the first term $a = 17$ $[a = S_1 = 1(15+2\times1) = 17]$
	M1	For the second term. Uses the given summation formula to form a linear equation in a and d for a minimally acceptable response of $k = 2a+d$ where k is a positive integer.
	A1	For the correct linear equation $38 = 2a+d$
	A1	For the correct value of $d = 4$
	ALT 1	
	B1	For the first term $a = 17$
	M1	For using a correct summation formula $n(15+2n) = \frac{n}{2}(2a+[n-1]d) \Rightarrow 30+2n = 2a-d+nd$ and equates coefficients
	A1	For equating coefficients of n $4n = dn \Rightarrow d = \ldots$ and $30 = 2a-4 \Rightarrow a = \ldots$
		For the correct value of $d = 4$

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	ALT 2	
	B1	For the first term $a = 17$
	M1	Uses two values of n to set up a pair of simultaneous equations.e.g. $S_4 = 4(15+2\times4) = 92$ and $92 = \frac{4}{2}(2a+3d) \Rightarrow 46 = 2a+3d$ $S_5 = 5(15+2\times5) = 125$ and $125 = \frac{5}{2}(2a+4d) \Rightarrow 50 = 2a+4d$
	A1	Attempts to solve the pair of equations
	A1	$d = 4$
(b)	M1	For using the correct nth term formula with **their a and their d** $U_{20} = '17'+19\times'4' = ...$
	A1	For the correct 20th term = 93
(c)	M1	Uses the given summation formula with the correct substitution $2p(15+4p)-2p(15+2p) = 1+(p-1)(13+2p)$
	A1	For achieving the correct 3TQ $2p^2-11p+12 = 0$
	ALT	
	M1	Uses the summation formula: **Follow through their a and d** $S_{2p} = \frac{2p}{2}(2\times17+(2p-1)4) = p(30+8p)$ $2S_p = 2\times\frac{p}{2}(2\times17+(p-1)4) = p(30+4p)$ $S_{p-1} = \frac{p-1}{2}(2\times17+(2[p-1]-1)4) = (p-1)(13+2p)$ For a correct substitution into the given expression $p(30+8p)-p(30+4p) = 1+(p-1)(13+2p)$ oe
	A1	For achieving the correct 3TQ $2p^2-11p+12 = 0$
	M1	For attempting to solve their 3TQ (provided it is a 3TQ) by any valid method. $2p^2-11p+12 = (2p-3)(p-4) = 0 \Rightarrow p = ..., ...$
	A1	For $p = 4$ If they give both roots of their 3TQ as an answer without rejecting $p = 1.5$ A0

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Question Number	Scheme	Marks
7(a)	$x^2-9x+14=\left(x-\frac{9}{2}\right)^2+14-\frac{81}{4}=\left(x-\frac{9}{2}\right)^2-\frac{25}{4}$	M1
	$a=-\frac{9}{2},\ b=-\frac{25}{4}$ oe	A1 (2)
(b)	(i) least value of $\mathrm{f}(x)=-\frac{25}{4}$	B1ft
	(ii) least value when $x=\frac{9}{2}$	B1ft (2)
(c)	$x+5=x^2-9x+14$	M1
	$x^2-10x+9=0\Rightarrow(x-9)(x-1)=0$	M1
	Points are $(9,14)$ $(1,6)$	A1A1 (4)
(d)	Area $\int_1^9\left((x+5)-\left(x^2-9x+14\right)\right)\mathrm{d}x=\int_1^9\left(-x^2+10x-9\right)\mathrm{d}x$	M1
	$=\left[-\frac{x^3}{3}+5x^2-9x\right]_1^9$	M1A1
	$=(-243+405-81)-\left(-\frac{1}{3}+5-9\right)=85\frac{1}{3}$	M1A1 (5)
		[13]

Part	Mark	Notes
(a)	M1	For attempting to complete the square to achieve as a minimum $x^2-9x+14=\left(x\pm\frac{9}{2}\right)^2+14-k$ where k is a constant
	A1	For the correct expression $x^2-9x+14=\left(x-\frac{9}{2}\right)^2-\frac{25}{4}$ or $a=-\frac{9}{2},\ b=-\frac{25}{4}$ oe stated
(b)(i)	B1ft	For the correct value $\mathrm{f}(x)=-\frac{25}{4}$ follow through their value of '$-\frac{25}{4}$'
(ii)	B1ft	For the correct value of $x=\frac{9}{2}$ provided they have $\left(x-\frac{9}{2}\right)^2$ …in part (a). Follow through their value of '$\frac{9}{2}$'

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(c)	M1	For equating the equation of the line with the equation of C $x+5=x^2-9x+14\Rightarrow x^2-10x+9=0$ **and** attempting to form a 3TQ
	M1	Attempts to solve their 3TQ by any method, provided it is the result of equating the line with C $x^2-10x+9=0\Rightarrow(x-9)(x-1)=0$
	A1	For the correct coordinates of **either** $(9,14)$ or $(1,6)$
	A1	For both correct pairs of coordinates $(9,14)$ and $(1,6)$
(d)	M1	For a correct expression for the required area with both limits correct. (ft their limits from (c)) Award this mark if they have 'curve – line' but otherwise correct. $\int_1^9\left((x+5)-\left(x^2-9x+14\right)\right)\mathrm{d}x=\left[\int_1^9\left(-x^2+10x-9\right)\mathrm{d}x\right]$, accept $\int_1^9\left(x^2-10x+9\right)\mathrm{d}x$ **OR** Area under the trapezium – curve $\frac{1}{2}\times8\times(6+14)-\int_1^9\left(x^2-9x+14\right)\mathrm{d}x$
	M1	For attempting to integrate the equation for the combined expression or the curve only.
	A1	For the correct integrated expression for required area. Ignore limits for this mark – even if they are absent altogether. Area = $=\left[-\frac{x^3}{3}+5x^2-9x\right]_1^9$ accept $\left[\frac{x^3}{3}-5x^2+9x\right]_1^9$ **OR** $\frac{1}{2}\times8\times(6+14)-\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9$ **OR** $\left(\frac{x^2}{2}+5x\right)_1^9-\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9$ or $\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9-\left(\frac{x^2}{2}+5x\right)_1^9$
	M1	For substituting their limits (x – coordinates from part(c)) into their integrated expression. $=(-243+405-81)-\left(-\frac{1}{3}+5-9\right)=\ldots$ **OR** $80-\left[\left(\frac{9^3}{3}-\frac{9\times9^2}{2}+14\times9\right)-\left(\frac{1^3}{3}-\frac{9\times1^2}{2}+14\times1\right)\right]=\ldots$ **OR** $\left[\left(\frac{9^2}{2}+5\times9\right)-\left(\frac{1^2}{2}+5\times1\right)\right]-\left[\left(\frac{9^3}{3}-\frac{9\times9^2}{2}+14\times9\right)-\left(\frac{1^3}{3}-\frac{9\times1^2}{2}+14\times1\right)\right]=\ldots$
	A1	For the correct area of $85\frac{1}{3}$ or $\frac{256}{3}$ If they get a value of $-85\frac{1}{3}$ they must give a **final** value of $85\frac{1}{3}$ for this mark.

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 2 of 22 · Question 3 part a · 3 marks

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Question Number	Scheme	Marks
8(a)	$2xy+5y=e^x \quad y=\dfrac{e^x}{(2x+5)}$	
	$\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$	M1A1A1
	$\dfrac{dy}{dx}=\dfrac{e^x}{(2x+5)}\times\dfrac{(2x+5-2)}{(2x+5)}=\dfrac{y(2x+3)}{(2x+5)}$ *	M1A1 (5)
(b)	$x=0\Rightarrow \dfrac{dy}{dx}=\dfrac{5-2}{5^2}=\dfrac{3}{25}$	M1A1 (2)
ALT	$x=0\Rightarrow y=\dfrac{1}{5},\ \dfrac{dy}{dx}=\dfrac{1}{5}\times\dfrac{3}{5}=\dfrac{3}{25}$	
(c)	$x=0\Rightarrow y=\dfrac{e^0}{(2\times0+5)}=\dfrac{1}{5}$	M1(Award if seen in (b) and **used** in (c))
	$y-\dfrac{1}{5}=-\dfrac{25}{3}x$	M1
	$125x+15y-3=0$	A1 (3) [10]

Part	Mark	Notes
(a)		$2xy+5y=e^x \Rightarrow y=\dfrac{e^x}{(2x+5)}$
	M1	For attempting Quotient Rule • Both terms must be differentiated correctly $e^x\Rightarrow e^x \quad 2x+5\Rightarrow 2$ • There must be two terms subtracted in the numerator either way around • The denominator must the denominator squared. $\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$
	A1	For $e^x(2x+5)$ or $2e^x$
	A1	For a fully correct differentiated expression. $\dfrac{dy}{dx}=\dfrac{e^x(2x+5)-2e^x}{(2x+5)^2}$

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 2 of 22 · Question 3 part a · 3 marks

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	M1	Subs in $y=\frac{e^x}{(2x+5)}$ as a common factor $\frac{dy}{dx}=\frac{e^x}{(2x+5)}\times\frac{(2x+5-2)}{(2x+5)}=\frac{y(2x+5-2)}{(2x+5)}$	Subs in $e^x=y(2x+5)$ and factorises $\frac{dy}{dx}=\frac{y(2x+5)(2x+5)-2y(2x+5)}{(2x+5)^2}$
	A1	For the correct answer with no errors. **Note this is a given answer.** $\frac{dy}{dx}=\frac{y(2x+3)}{(2x+5)}$	
	ALT – uses implicit differentiation on $2xy+5y=e^x$		
	M1	$2\left(y+x\frac{dy}{dx}\right)+5\frac{dy}{dx}=e^x$	
	A1	Takes out $\frac{dy}{dx}$ as a common factor $\frac{dy}{dx}(2x+5)=e^x-2y$	
	A1	For a fully correct differentiated expression as below. $\frac{dy}{dx}=\frac{e^x-2y}{(2x+5)}$	
	M1	For separating the fraction, taking out y as a common factor and attempting to form a single fraction $\frac{dy}{dx}=\frac{e^x}{(2x+5)}-\frac{2y}{(2x+5)}=\frac{y(2x+5)}{(2x+5)}-\frac{2y}{(2x+5)}=\frac{y(2x+5-2)}{2x+5}$	
	A1	For the correct answer with no errors. $\frac{dy}{dx}=\frac{y(2x+3)}{(2x+5)}$	
(b)	M1	For substituting $x=0$ into $\frac{dy}{dx}=\frac{e^x(2x+5)-2e^x}{(2x+5)^2}=\frac{e^0(2\times0+5)-2e^0}{(2\times0+5)^2}=...$	
	A1	For the correct value of $\frac{dy}{dx}=\frac{3}{25}$	
	ALT		
	M1	When $x=0\Rightarrow y=\frac{1}{5}$, $\frac{dy}{dx}=\frac{1}{5}\times\frac{3}{5}=...$	
	A1	For the correct value of $\frac{dy}{dx}=\frac{3}{25}$	
(c)	M1	$x=0\Rightarrow y=\frac{e^0}{(2\times0+5)}=\frac{1}{5}$ $x=0\Rightarrow y=\frac{1}{5}$ Award if seen in (b) and **used** in (c) **This is a B mark in Epen.**	
	M1	Inverts the gradient found in (b) and forms equation of the normal. ft their value of y $y-\frac{1}{5}=-\frac{25}{3}x$	
	A1	Equation of line is given in the required form. $125x+15y-3=0$	

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 2 of 22 · Question 3 part a · 3 marks

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Question Number	Scheme	Marks
9(a)	$\cos ADB = \dfrac{6^2 + x^2 - 12^2}{2\times 6x} = \dfrac{x^2 - 108}{12x}$	M1A1 (2)
(b)	$\cos BDC = \dfrac{6^2 + x^2 - 6^2}{2\times 6x} = \dfrac{x^2}{12x}$	B1
	$ADB = 180 - BDC \Rightarrow -\dfrac{x^2}{12x} = \dfrac{x^2 - 108}{12x}$	M1A1
	$2x^2 = 108 \Rightarrow x = 3\sqrt{6}$	
	$AC = 6\sqrt{6}$	A1 (4)
(c)	$\dfrac{\sin(\theta^\circ + \phi^\circ)}{2x} = \dfrac{\sin BCD}{12} \Rightarrow \dfrac{\sin(\theta^\circ + \phi^\circ)}{x} = \dfrac{\sin BCD}{6}$	M1A1
	$\dfrac{\sin\phi^\circ}{x} = \dfrac{\sin BCD}{6}$	M1
	$\therefore \sin\phi^\circ = \sin(\theta^\circ + \phi^\circ)$	A1 (4)
(d)	$\sin(\theta^\circ + \phi^\circ) = \sin\phi^\circ \Rightarrow (\theta + \phi) = 180 - \phi$ (or ϕ or $360 + \phi$ (not possible))	M1
	$\therefore \theta = 180 - 2\phi$	A1 (2) [12]

Part	Mark	Notes
(a)	M1	For using a correct cosine rule $\cos ADB = \dfrac{6^2 + x^2 - 12^2}{2\times 6x} = \dfrac{x^2 - 108}{12x}$ or $12^2 = x^2 + 6^2 - 2\times 6\times x\cos ADB$
	A1	Simplifies to $\cos ADB = \dfrac{x^2 - 108}{12x}$
(b)	B1	For the correct expression $\cos BDC = \dfrac{6^2 + x^2 - 6^2}{2\times 6x} = \dfrac{x^2}{12x}$
	M1	$\cos BDC = -\cos ADB$ and $\cos BDC = -\dfrac{x^2}{12x}$ so $-\dfrac{x^2}{12x} = \dfrac{x^2 - 108}{12x}$ and attempts to solve
	ALT 1 – uses triangles *BAD* and *BAC*	
	B1	For **both** of the following correct expressions for $\cos BAD$ and $\cos BAC$: $\cos BAD = \dfrac{12^2 + x^2 - 6^2}{2\times 12\times x}$ and $\cos BAC = \dfrac{12^2 + (2x)^2 - 6^2}{2\times 12\times 2x}$
	M1	$\angle BAC = \angle BAD$ so equates their two expressions $\dfrac{12^2 + x^2 - 6^2}{2\times 12\times x} = \dfrac{12^2 + (2x)^2 - 6^2}{2\times 12\times 2x} \Rightarrow \dfrac{108 + x^2}{24x} = \dfrac{108 + 4x^2}{48x}$ and attempts to solve
	ALT 2 – uses triangles *BCD* and *BCA*	

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 2 of 22 · Question 3 part a · 3 marks

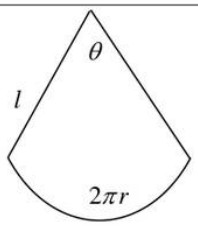
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	B1	For **both** of the following correct expressions for cos BAD and cos BAC: $\cos BCD = \frac{6^2+x^2-6^2}{2\times6\times x}$ and $\cos BCA = \frac{6^2+(2x)^2-12^2}{2\times6\times2x}$
	M1	$\frac{6^2+x^2-6^2}{2\times6\times x} = \frac{6^2+(2x)^2-12^2}{2\times6\times2x} \Rightarrow x^2 = 2x^2-54$ and attempts to solve
	Final A marks for all three methods	
	A1	For the correct value of $x = 3\sqrt{6}$
	A1	For $AC = 6\sqrt{6}$
(c)	M1	Uses sine rule on triangle ABC: $\frac{\sin(\theta^\circ+\phi^\circ)}{2x} = \frac{\sin BCD}{12} \Rightarrow \frac{\sin(\theta^\circ+\phi^\circ)}{x} = \frac{\sin BCD}{6}$
	A1	Achieves the correct expression for $\sin(\theta^\circ+\phi^\circ) = \frac{x\sin BCD}{6}$
	M1	Uses sine rule on triangle BDC: $\frac{\sin\phi^\circ}{x} = \frac{\sin BCD}{6} \Rightarrow \left(\sin\phi^\circ = \frac{x\sin BCD}{6}\right)$
	A1	Shows that $\sin\phi^\circ = \sin(\theta^\circ+\phi^\circ)$ with no errors
	ALT 1 – Uses exact values for the trigonometric ratios and the expansion for sin $(A + B)$	
	M1	Finds $\cos\theta = \frac{7}{8} \Rightarrow \sin\theta = \frac{\sqrt{15}}{8}$ **or** $\cos\phi = \frac{1}{4} \Rightarrow \sin\phi = \frac{\sqrt{15}}{4}$ Accept $\theta = 28.95...^\circ$ **or** $\phi = 75.52...^\circ \Rightarrow \sin\phi = 0.968...$
	A1	Finds $\cos\theta = \frac{7}{8} \Rightarrow \sin\theta = \frac{\sqrt{15}}{8}$ **and** $\cos\phi = \frac{1}{4} \Rightarrow \sin\phi = \frac{\sqrt{15}}{4}$ Accept $\theta = 28.95...^\circ$ **and** $\phi = 75.52...^\circ \Rightarrow \sin\phi = 0.968...$
	M1	Expands $\sin(\theta+\phi) = \frac{\sqrt{15}}{8}\times\frac{1}{4} + \frac{\sqrt{15}}{4}\times\frac{7}{8} = \left(\frac{\sqrt{15}}{4}\right)$ Or $\sin(\theta+\phi)^\circ = \sin(28.95^\circ)\cos(75.52^\circ)+\sin(75.52^\circ)\cos(28.95^\circ) = 0.968... = \sin\phi^\circ$
	A1	Shows that $\sin(\theta+\phi) = \frac{\sqrt{15}}{4}$ and $\sin\phi = \frac{\sqrt{15}}{4}$ so $\sin(\theta+\phi) = \sin\phi$ with no errors. If they use approximate values for sin θ and ϕ withhold this final mark so A0
	ALT 2 – Uses $\angle BCD = 52.2...^\circ$	
	M1	Finds $\angle BCD = 52.2...^\circ$ using cosine rule and applies sine rule on triangle ABC $\frac{\sin(\theta^\circ+\phi^\circ)}{6\sqrt{6}} = \frac{\sin 52.2^\circ}{12}$
	A1	Shows that $\sin(\theta^\circ+\phi^\circ) = 0.968...$
	M1	Uses sine rule on triangle BD: $\frac{\sin\phi^\circ}{3\sqrt{6}} = \frac{\sin 52.2^\circ}{6} \Rightarrow \sin\phi^\circ = 0.968... = \sin(\theta^\circ+\phi^\circ)$
	A1	If they use an approximate value for angle BCD withhold this final mark so A0
(d)	M1	For writing $\sin\phi^\circ = \sin(180-\phi)^\circ \Rightarrow \theta^\circ+\phi^\circ = 180^\circ-\phi^\circ$
	A1	For rearranging $\theta^\circ+\phi^\circ = 180^\circ-\phi^\circ$ to achieve $\therefore \theta = 180-2\phi$ This is a show question and there must be no errors here.

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 2 of 22 · Question 3 part a · 3 marks

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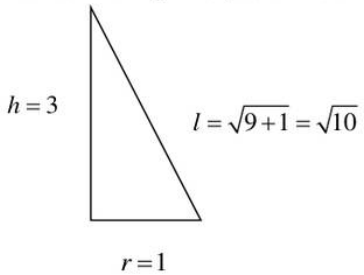
Question Number	Scheme	Marks
10(a)	Circumference of base $= 2\pi r$ 	B1
	$l\theta = 2\pi r \Rightarrow \theta = \dfrac{2\pi r}{l}$	B1
	$A = \dfrac{1}{2}l^2\theta = \dfrac{1}{2}l^2\dfrac{2\pi r}{l} = \pi rl$	M1A1 (4)
(b)	$A = \pi rl$	
	$l = r\sqrt{10} \Rightarrow A = \pi r^2\sqrt{10}$	B1
	$\dfrac{dA}{dr} = 2\pi r\sqrt{10} \Rightarrow k = 2\sqrt{10}$	M1A1 (3)
(c)	$\dfrac{dV}{dt} = 1.5\ \left(\text{cm}^3/\text{s}\right)$	B1
	$\dfrac{dA}{dt} = \dfrac{dA}{dr} \times \dfrac{dr}{dV} \times \dfrac{dV}{dt}$	M1
	$V = \dfrac{1}{3}\pi r^2 h = \dfrac{1}{3}\pi r^2 \times 3r = \pi r^3$	B1
	$\dfrac{dV}{dr} = 3\pi r^2$	B1ft
	$\therefore \dfrac{dA}{dt} = 2\pi \times 8\sqrt{10} \times 1.5 \times \dfrac{1}{3 \times 64\pi} = 0.3952... = 0.395\ \text{cm}^2/\text{s}$	A1 (5)
		[12]

Part	Mark	Notes	
(a)	B1	For the circ of the base $L = 2\pi r$	
	B1	$R = l$ and $l = r\theta$ Therefore $l\theta = 2\pi r \Rightarrow \theta = \dfrac{2\pi r}{l}$	$R = l$
	M1	$A = \dfrac{1}{2}l^2\theta$ and substituting their expression for θ to give $A = \dfrac{1}{2}l^2\theta = \dfrac{1}{2}l^2\dfrac{2\pi r}{l}$	Uses the formula $A = \dfrac{1}{2}RL$ $A = \dfrac{1}{2} \times l \times 2\pi r \Rightarrow \left(A = \pi rl\right)$
	A1	For the required expression for A, $A = \pi rl$ with no errors.	

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 2 of 22 · Question 3 part a · 3 marks

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(b)	B1	For finding that the slant height is $\sqrt{10}$ times the radius of the cone  So $l=r\sqrt{10}$
	M1	Substitutes $l=r\sqrt{10}$ into the given expression $A=\pi rl$ and differentiates their resulting expression to find $\frac{dA}{dr}$ $A=\pi r^2\sqrt{10}$ so $\frac{dA}{dr}=2\pi r\sqrt{10}$
	A1	Therefore $k=2\sqrt{10}$
(c)	B1	States $\frac{dV}{dt}=1.5\ \left(\text{cm}^3/\text{s}\right)$ Award if it seen explicitly in (b) **and used** in (c)
	M1	States (or uses) a correct chain rule $\frac{dA}{dt}=\frac{dA}{dr}\times\frac{dr}{dV}\times\frac{dV}{dt}$
	B1	For finding the volume of a cone in terms of r only $V=\frac{1}{3}\pi r^2h=\frac{1}{3}\pi r^2\times 3r=\pi r^3$
	B1ft	Differentiates their expression for the volume of a cone provided it is in terms of V and r only. $\frac{dV}{dr}=3\pi r^2$
	A1	For combining all required terms into their chain rule and evaluating to 3 significant figures, $\frac{dA}{dt}=2\pi\times 8\sqrt{10}\times 1.5\times\frac{1}{3\times 64\pi}=0.3952...=0.395\ \text{cm}^2/\text{s}$
	ALT – in terms of h	
	B1	States $\frac{dV}{dt}=1.5\ \left(\text{cm}^3/\text{s}\right)$ Award if it seen explicitly in (b) **and used** in (c)
	M1	States (or uses) a correct chain rule $\frac{dA}{dt}=\frac{dA}{dh}\times\frac{dh}{dt}$ and $\frac{dV}{dt}=\frac{dV}{dh}\times\frac{dh}{dt}$
	B1	For finding the area and volume of a cone in terms of h $r=\frac{h}{3}$, $A=\pi r^2\sqrt{10}\Rightarrow A=\frac{\sqrt{10}}{9}\pi h^2$ and $V=\frac{1}{3}\pi r^2h\Rightarrow V=\frac{1}{27}\pi h^3$
	B1ft	Differentiates their expressions for the area and volume of a cone provided they are both in terms of h only. $\frac{dA}{dh}=\frac{2\sqrt{10}}{9}\pi h$ and $\frac{dV}{dh}=\frac{3}{27}\pi h^2$
	A1	Combines the required terms into their chain rules and evaluating to 3 significant figures $\frac{dA}{dt}=0.395\ 102$

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 2 of 22 · Question 3 part a · 3 marks

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Question Number	Scheme	Marks
11(a)	(i) $\overrightarrow{AC} = -\mathbf{a} + \frac{1}{4}\mathbf{b}$	B1
	(ii) $\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC} = \mathbf{a} + \frac{1}{2}\left(-\mathbf{a} + \frac{1}{4}\mathbf{b}\right) = \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b}$	M1A1
	(iii) $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OD} = -\mathbf{b} + \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b} = \frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}$	M1A1 (5)
(b)	$\overrightarrow{BE} = k\overrightarrow{BD} = k\left(\frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}\right)$	M1
	$\overrightarrow{BE} = -\mathbf{b} + \overrightarrow{OE} = -\mathbf{b} + \lambda\mathbf{a}$	M1
	$\frac{7}{8}k = 1 \Rightarrow k = \frac{8}{7}$	M1
	$\frac{k}{2} = \lambda \Rightarrow \lambda = \frac{4}{7}$	A1 (4)
(c)	$\triangle OAC = \frac{1}{4}\triangle OAB$	M1
	$\triangle OEB = "\frac{4}{7}"\triangle OAB$	A1 ft
	$\frac{\triangle OAC}{\triangle OEB} = \frac{1}{4} \times "\frac{7}{4}" = \frac{7}{16}$	M1
	$\mu = \frac{7}{16}$	A1 (4)
		[13]

Part	Mark	Notes
(a)(i)	B1	For the correct vector $\overrightarrow{AC} = -\mathbf{a} + \frac{1}{4}\mathbf{b}$
(ii)	M1	For the correct vector statement $\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC}$ or $\overrightarrow{OD} = \overrightarrow{OC} + \frac{1}{2}\overrightarrow{CA}$
	A1	For the correct **simplified** vector $\overrightarrow{OD} = \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b}$
(iii)	M1	For the correct vector statement $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OD}$ or $\overrightarrow{BD} = \overrightarrow{BC} + \overrightarrow{CD}$
	A1	For the correct **simplified** vector $\overrightarrow{BD} = \frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}$

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 2 of 22 · Question 3 part a · 3 marks

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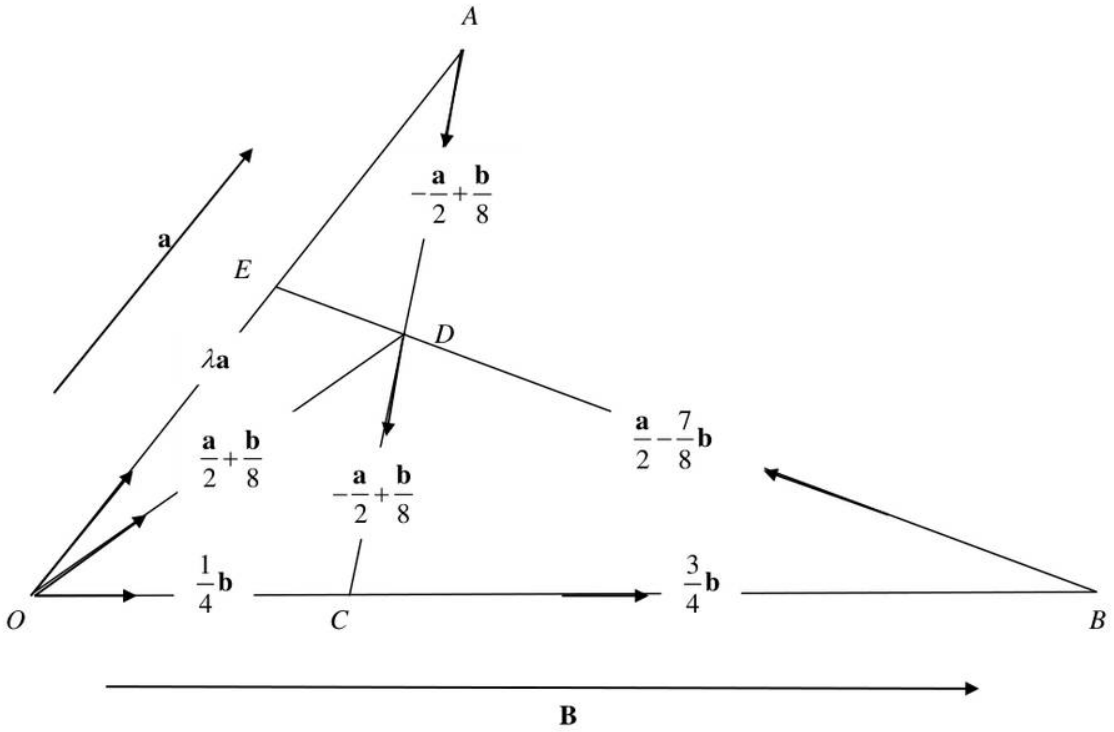
In parts (b) and (c) you must follow through the vectors they have found in part (a)		
(b)	M1	States or implies two paths that will lead to a solution. For example: using triangle OEB • $\overrightarrow{BE}=k\overrightarrow{BD}$ and $\overrightarrow{BE}=\overrightarrow{BO}+\overrightarrow{OE}$ **or** using triangle OED • $\overrightarrow{OE}=\lambda\mathbf{a}$ and $\overrightarrow{OE}=\overrightarrow{OD}+\overrightarrow{DE}$ **This is a B mark in Epen**
	M1	For writing their paths as vectors in terms of **a**, **b** λ **and another constant** (e.g. k or μ) For example: using triangle OEB • $\overrightarrow{BE}=k\overrightarrow{BD}=k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$ and $\overrightarrow{BE}=-\mathbf{b}+\lambda\mathbf{a}$ **or** using triangle OED • $\overrightarrow{OE}=\lambda\mathbf{a}$ and $\overrightarrow{OE}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(-\mathbf{b}+\lambda\mathbf{a}\right)$ OR $\overrightarrow{OE}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$
	dM1	Equates coefficients of their **two** expressions and attempts to find the value of λ: In triangle OEB • $k\frac{1}{2}\mathbf{a}-k\frac{7}{8}\mathbf{b}=\lambda\mathbf{a}-\mathbf{b}$ $\Rightarrow k\frac{7}{8}=1\Rightarrow k=\frac{8}{7}$ and $\frac{k}{2}=\lambda\Rightarrow\lambda=...$ or in triangle OED, there are two possible expressions for $\overrightarrow{DE}$ • $\lambda\mathbf{a}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(-\mathbf{b}+\lambda\mathbf{a}\right)$ $\Rightarrow k=\frac{1}{8}$ and $\lambda=\frac{1}{2}+\frac{1}{8}\lambda$ $\Rightarrow\lambda=...$ Or $\lambda\mathbf{a}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$ $\Rightarrow\frac{7}{8}k=\frac{1}{8}\Rightarrow k=\frac{1}{7}$ and $\lambda=\frac{1}{2}+\frac{1}{7}k\Rightarrow\lambda...$
	A1	For the correct value of λ $\lambda=\frac{4}{7}$
(c)	M1	For stating that $\triangle OAC=\frac{1}{4}\triangle OAB$ or $4\triangle OAC=\triangle OAB$ **1**
	M1	For stating $\triangle OEB="\lambda"\triangle OAB\Rightarrow\triangle OEB="\frac{4}{7}"\triangle OAB$ **2** **This is an A mark in Epen**
	M1	For dividing **1** by **2** $=\frac{\triangle OAC}{\triangle OEB}=\frac{1}{4}\times"\frac{7}{4}"=...$
	A1	For $\mu=\frac{7}{16}$

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 2 of 22 · Question 3 part a · 3 marks

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USEFUL SKETCH



Type MS · Lengths in a 3D solid using right-angled triangles

Answer 3 of 22 · Question 8 part b · 3 marks

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8(a)	$\tan 30° = \frac{VP}{6}$ where P is the centre of the base $VP = 6\tan 30° = 2\sqrt{3}$ *	M1 A1*cso [2]
(b)	$AC = \sqrt{12^2 + 12^2} = 12\sqrt{2}$ or $AC = \frac{12}{\sin 45}$ or $AC = \frac{12}{\cos 45}$ $VA^2 = \left(2\sqrt{3}\right)^2 + \left(6\sqrt{2}\right)^2 (= 84)$ $VA = 2\sqrt{21}$ (cm) oee	M1 oe M1 oe A1 [3]
(c)	$VX = \sqrt{\left('2\sqrt{21}'\right)^2 - 6^2} \left(= 4\sqrt{3}\right)$ or $VX = \frac{6}{\cos 30}$ or $VX = \sqrt{6^2 + \left(2\sqrt{3}\right)^2}$ where X is the midpoint of the side AD Area $= \frac{1}{2} \times 4\sqrt{3} \times 12 = 24\sqrt{3}$ (cm^2) oee	M1 M1A1 [3]
(d)	$'24\sqrt{3}' = \frac{1}{2} \times '2\sqrt{21}' \times DY$ where Y is the foot of the perpendicular from D to VA $DY = \frac{24\sqrt{7}}{7}$ oee	M1 A1 [2]
(e)	$\cos\theta = \frac{\left(,\frac{24\sqrt{7}}{7},\right)^2 + \left(,\frac{24\sqrt{7}}{7},\right)^2 - \left('12\sqrt{2}'\right)^2}{2 \times ,\frac{24\sqrt{7}}{7}, \times ,\frac{24\sqrt{7}}{7},} = -\frac{3}{4}$ $\theta = 138.6°$	M1A1 A1 [3]
		Total is 13 marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 3 of 22 · Question 8 part b · 3 marks

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Part	Mark	Additional Guidance
(a)	M1	For forming a correct equation using trigonometry. e.g. $\tan 60° = \frac{6}{VP}$ or $\frac{h}{\sin 30°} = \frac{6}{\sin 60°}$ (If attempted by ratio, send to review)
	A1*cso	For **rearranging** to show the required result. Minimum need to see one intermediate step from their equation to the answer ,if the height is not the subject of their equation: e.g. $\tan 30° = \frac{VP}{6} \Rightarrow VP = 6\tan 30° = 2\sqrt{3}$ is M1A1 $VP = 6\tan 30° = 2\sqrt{3}$ is M1A1 $\tan 30° = \frac{VP}{6} \Rightarrow \frac{\sqrt{3}}{3} = \frac{VP}{6} \Rightarrow VP = 2\sqrt{3}$ is M1A1 $\tan 30° = \frac{VP}{6} \Rightarrow VP = 2\sqrt{3}$ is M1A0
(b)	M1oe	Correct use of Pythagoras' theorem or trigonometry to obtain the length/half length of the diagonal of the base AC or $\frac{1}{2}AC$ Or other valid method to find VM, (Let M to be the midpoint of AB) e.g.: $VM = \frac{6}{\cos 30}\left(= 4\sqrt{3}\right)$ or $VM = \sqrt{6^2 + \left(2\sqrt{3}\right)^2}\ \left(= 4\sqrt{3}\right)$
	M1oe	Use Pythagoras' theorem to obtain a numerical value for VA / VA^2 Any correct method. e.g. $VA^2 = \left(4\sqrt{3}\right)^2 + (6)^2$
	A1	Correct exact length $2\sqrt{21}$ or $\sqrt{84}$ oee (or exact equivalent)
(c)	M1	Correct use of Pythagoras' theorem or trigonometry to obtain the length of VX, which is the same as VM, could be found in part b Or correct use of cosine rule (ft their VA) to find the value of cosine value of any angle in triangle VAD e.g. $AD^2 = VA^2 + VA^2 - 2VA \times VA \times \cos\angle AVD$ $12^2 = \left('2\sqrt{21}'\right)^2 + \left('2\sqrt{21}'\right)^2 - 2 \times '2\sqrt{21}' \times '2\sqrt{21}' \times \cos\angle AVD$ $\left\{\Rightarrow \cos\angle AVD = \frac{1}{7}\right\}$
	M1	Correct method to find the area of triangle VAD (Do not have to use exact values) Accept any correct valid method. e.g. $\left\{\cos\angle AVD = \frac{1}{7} \Rightarrow \sin\angle AVD = \frac{4\sqrt{3}}{7}\right\} \Rightarrow A = \frac{1}{2} \times '2\sqrt{21}' \times '2\sqrt{21}' \times '\frac{4\sqrt{3}}{7}' = 24\sqrt{3}$
	A1	Correct **exact** value for area
(d)	M1	Use of area of triangle with VA as the base to form an equation for DY Or other valid method: e.g. using trigonometry ratio: $\sin\angle VAD = \frac{'4\sqrt{3}'}{'2\sqrt{21}'} = \frac{DY}{12}$

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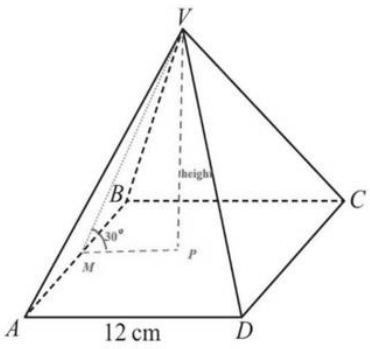
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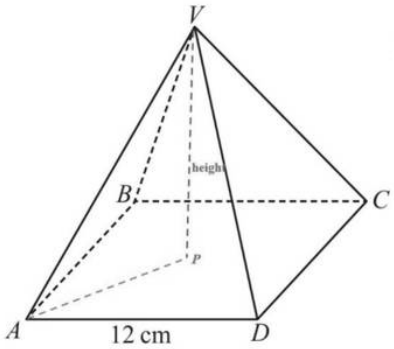
	A1	Correct exact length $DY = \frac{24\sqrt{7}}{7}$ oee, e.g. $DY = \frac{8\sqrt{63}}{7}$ or $24\sqrt{\frac{1}{7}}$
(e)	M1	Correct use of cosine rule: $\cos\theta = \frac{"DY"^2 + BY"^2 - "BD"^2}{2\times"DY"\times"BY"}$ with their values Or other valid method, e.g. $\sin\frac{\theta}{2} = \frac{\frac{"BD"}{2}}{"DY"} = \frac{"6\sqrt{2}"}{\frac{"24\sqrt{7}"}{7}}$
	A1	Obtains $\cos\theta = -\frac{3}{4}$ or $\sin\frac{\theta}{2} = \frac{\sqrt{14}}{4}$
	A1	Correct angle found

Useful diagrams:

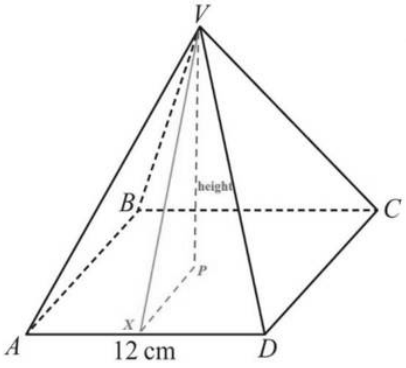
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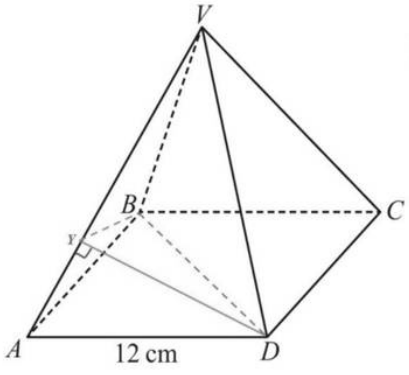
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Question number	Scheme	Marks

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8 (a)	$BN = 10\cos 60^\circ$ $= 5$ (cm)*	M1 A1cso [2]
(b)	$FN^2 = 10^2 - 5^2 = 75$ or $FN = 10\sin 60 = 5\sqrt{3}$ or $FN = 5\tan 60$ $NE = \sqrt{12^2 + 75} = 14.7986... \approx 14.8$ (cm)	M1 M1A1 [3]
(c)	Let R be the midpoint of FN and P be the point on AD that corresponds to point N on BC. $\angle RPN = \tan^{-1}\left(\dfrac{\sqrt{75}/2}{12}\right) = 19.84167...^\circ \approx 19.8^\circ$	M1A1 [2]
(d)	Let the perpendicular from Y to line BC meet BC at Z $AY = \sqrt{AZ^2 + ZY^2} = \sqrt{12^2 + 6.5^2 + \left(\dfrac{\sqrt{75}}{2}\right)^2} = \sqrt{205}$ $FC = \sqrt{FN^2 + NC^2} = \sqrt{75 + 3^2} = \sqrt{84} \Rightarrow FY = \sqrt{21}$ So $EY = \sqrt{12^2 + 21} = \sqrt{165}$ $\angle AYE = \cos^{-1}\left(\dfrac{205 + 165 - 100}{2 \times \sqrt{205} \times \sqrt{165}}\right) = 42.7745...^\circ \approx 42.8^\circ$	M1A1 M1 M1 M1A1 [6]
		Total 13 marks

(a)	M1	Use cosine or sine in triangle BNF, or any complete method. $BN = 10\cos 60^\circ$ or $\cos 60^\circ = \dfrac{BN}{10}$ or $\dfrac{BN}{\sin 30} = \dfrac{10}{\sin 90}$ or $BN = 10\sin 30$ or $FN = 10\sin 60 = 5\sqrt{3}$ and $BN^2 = 10^2 - (5\sqrt{3})^2$ (may use any single letter instead of BN)
	A1 cso	$BN = 5$ (cm) Correct answer from correct working. (Answer given.) Must show a calculation in the form $BN = \ldots$ e.g. $BN = 10\cos 60$
(b)	M1	Use Pythag or trig to find a numerical expression for FN or FN^2 eg $FN^2 = 10^2 - 5^2$ $(= 75)$ or $FN = 10\sin 60$ $(= 5\sqrt{3} = 8.66...)$ or $FN = 5\tan 60$
	M1	Use Pythag with their FN to find a numerical expression for NE, eg $NE = \sqrt{12^2 + (5\sqrt{3})^2}$ $(= \sqrt{219})$
	A1	$NE = 14.8$ (cm) **given to 3SF**.
(c)	M1	Use tangent with their FN, eg $\tan RPN = \left(\dfrac{\frac{1}{2}\sqrt{75}}{12}\right)$ or any complete method.
	A1	19.8° **given to 1DP**.

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(d)	M1	Complete method to find AY eg $AY=\sqrt{(12^2+6.5^2)+\left(\frac{\sqrt{75}}{2}\right)^2}$ $(=\sqrt{205}=14.31...)$ (allow their FN)
	A1	$AY=\sqrt{205}$ $(=14.31\ldots)$ (may be implied by subsequent working)
	M1	Complete method to find FY eg $FC=\sqrt{75+3^2}\left(=\sqrt{84}\right)\Rightarrow FY=\frac{1}{2}\sqrt{84}$ $(=\sqrt{21})$ (allow their FN) eg $FC=\sqrt{8^2+10^2-2\times8\times10\cos60}\left(=\sqrt{84}\right)\Rightarrow FY=\frac{1}{2}\sqrt{84}$ $(=\sqrt{21}=4.58...)$
	M1	Method to find EY using their FY (correct or from a correct method) eg $EY=\sqrt{12^2+21}$ $(=\sqrt{165}=12.84...)$
	M1	Substitute their values for AY and EY into the cosine rule used in triangle AYE. $\cos AYE=\left(\frac{205+165-100}{2\times\sqrt{205}\times\sqrt{165}}\right)$ oe
	A1	$\angle AYE=42.8^\circ$ **given to 1DP**. Allow more than 1DP if 1DP was penalised in part (c).

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Question number	Scheme	Marks

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Answer 5 of 22 · Question 9 part b · 3 marks

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9 (a)	$\cos\alpha = \frac{3}{\sqrt{13}}$	B1 (1)
(b)	$h = \sqrt{17^2 - \left(9^2 + 12^2\right)} = 8$	M1 M1 A1 (3)
(c)	Let M be the midpoint of BC $\tan\theta^\circ = \frac{h}{OM} = \frac{8}{12} = \frac{2}{3}$ *	M1 A1 (2)
(d)	Let N be the midpoint of EM	
	$EM = \sqrt{8^2 + 12^2} = 4\sqrt{13}$	M1
	$NO = \sqrt{12^2 + \left(2\sqrt{13}\right)^2 - 2(12)\left(2\sqrt{13}\right)\left(\frac{3}{\sqrt{13}}\right)} \Rightarrow NO = 2\sqrt{13}$	M1
	Hence triangle ONM is isosceles $180 - 2\times 33.7 = 112.6^\circ$	M1 A1 (4)
		Total 10 marks

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Question number	Scheme	Marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 6 of 22 · Question 11 part b · 3 marks

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11(a)	$AC = \sqrt{(16^2 + 16^2)} = 16\sqrt{2}$	M1A1
ALT:	$AP^2 + PD^2 = 16^2 \Rightarrow AP = 8\sqrt{2}$	
	$VP = 8\sqrt{2}\tan 45 = 8\sqrt{2}$ (where P is the centre of the base) *	M1A1cso (4)
(b)	$VA^2 = \left(8\sqrt{2}\right)^2 + \left(8\sqrt{2}\right)^2 (= 256)$ or $VA = \dfrac{8\sqrt{2}}{\sin 45°}$	M1A1
	$VA = 16$ cm	A1 (3)
(c)	$DX^2 = 16^2 - 8^2$ where X is the foot of the perpendicular from D to VA	M1A1
	$DX = 8\sqrt{3}$	A1 (3)
(d)	$\tan\theta = \dfrac{8\sqrt{2}}{8}, \sin\theta = \dfrac{8\sqrt{2}}{8\sqrt{3}}, \cos\theta = \dfrac{8}{8\sqrt{3}}$	M1A1
	(or unsimplified if cosine or sine rule used) $\theta = 54.7°$	A1 (3)
(e)	$\cos\phi = \dfrac{\left(8\sqrt{3}\right)^2 + \left(8\sqrt{3}\right)^2 - \left(16\sqrt{2}\right)^2}{2\times 8\sqrt{3}\times 8\sqrt{3}} \left(= -\dfrac{1}{3}\right)$	M1A1
	$\phi = 109.5°$	A1 (3) [16]

(a)	
M1	Use Pythagoras (with a + sign) to obtain the length of the diagonal of the base. Or use Pythagoras with correct sign to obtain the half diagonal
A1	Correct length for the diagonal or half diagonal
M1	Use tan in ΔAPV, their AP and angle of 45° to obtain the height
A1cso	Correct answer with no errors in the working
	OR: State ΔAVP is isosceles, or shown the 2 correct angles – can be on a diagram M1
	$VP(= AP) = 8\sqrt{2}$ A1
	OR use any other complete valid method M1 Correct result A1
(b)	
M1	Use Pythagoras or trigonometry in ΔAPV or ΔAVC (or any other complete, valid method) to obtain a numerical expression for VA.
A1	Correct numbers in their choice of method.
A1	Correct length obtained.
	$AV = 16$ w/o working scores M1A1A1

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Question Number	Scheme	Marks
(c)		
M1	Use Pythagoras with a minus sign (seen or implied) or trigonometry in ΔADX OR any other complete valid method **NB**: triangle ΔADV is equilateral.	
A1	Correct numbers in their choice of method.	
A1	Correct **exact** length for the perpendicular.	
(d)		
M1	Identify the correct triangle needed with the required angle marked (may be on Figure 1). This may be shown explicitly or implied by their work that follows.	
A1	Reach one of $\tan\theta = \frac{8\sqrt{2}}{8}$, $\sin\theta = \frac{8\sqrt{2}}{8\sqrt{3}}$, $\cos\theta = \frac{8}{8\sqrt{3}}$ oe	
A1	Correct answer, **must be 1 dp.**	
(e)		
M1	Use cosine rule: $\cos\phi = \frac{"DX"^2 + "XB"^2 - "BD"^2}{2\times"DX"\times"XB"}$ (their values)	
A1	Correct numbers substituted, follow through their previous answers	
A1	Correct answer, **must be 1 dp unless already penalised in (d)**	
	Any other routes should be marked: M1 Correct, complete method (ie it must be possible to reach a value for the required angle) A1 Correct numbers substituted A1 Correct answer, **must be 1 dp unless already penalised in (d)**	

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Answer 7 of 22 · Question 4 part a · 2 marks

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4(a)	$FC = \sqrt{10^2 + 10^2 + 10^2} = \sqrt{300}$	M1A1 [2]
(b)	$\cos FCA = \dfrac{\sqrt{200}}{\sqrt{300}} \Rightarrow \angle FCA = 35.3^\circ$	M1A1 [2]
(c)	$CX = \dfrac{\sqrt{200}}{2} = \sqrt{50}$, $FX = \sqrt{50+100} = \sqrt{150}$ $\cos \angle FXC = \dfrac{'150'+'50'-'300'}{2\times\sqrt{'150'}\times\sqrt{'50'}} \Rightarrow \angle FXC = 125.2643...^\circ \Rightarrow$ awrt 125°	M1,M1A1 M1A1 [5]
	Total	**9 marks**

Part	Mark	Notes
(a)	M1	For using Pythagoras theorem or any appropriate trigonometry to find FC. $FC = \sqrt{10^2 + 10^2 + 10^2} = ...$ OR $AC = \sqrt{10^2 + 10^2} = \sqrt{200}$ $FC = \sqrt{10^2 + 200} = \sqrt{300}$
	A1	For the correct exact length of $FC = \sqrt{300}\left(=10\sqrt{3}\right)$
(b)	M1	For using any appropriate trigonometry to find the required angle. They must find a value for the award of this mark. $\cos \angle FCA = \dfrac{\sqrt{200}}{\sqrt{300}} \Rightarrow \angle FCA = \left(35.2643^\circ\right)$, $\sin \angle FCA = \dfrac{10}{\sqrt{300}} \Rightarrow \angle FCA = \left(35.2643^\circ\right)$, $\tan \angle FCA = \dfrac{10}{\sqrt{200}} \Rightarrow \angle FCA = \left(35.2643^\circ\right)$ Allow awrt 14.1 for $\sqrt{200}$ and awrt 17.3 for $\sqrt{300}$
	A1	$\angle FCA = 35.2643...^\circ \approx 35.3^\circ$ (awrt) Accept answers which round to 35.3°, but they must round to this value.

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(c)	M1	Let X be the midpoint of BH. For any method (Pythagoras or trigonometry) to find the length CX $CX = \frac{\sqrt{200}}{2} = \left(\sqrt{50} \text{ or } 5\sqrt{2}\right)$
	M1	For any method (Pythagoras or trigonometry) to find the length FX $FX = \sqrt{50+100} = \left(\sqrt{150} \text{ or } 5\sqrt{6}\right)$ or $FX = \sqrt{200-50} = \left(\sqrt{150} \text{ or } 5\sqrt{6}\right)$
	A1	For both correct lengths $\sqrt{50}$ and $\sqrt{150}$
	M1	For the correct cosine rule to find angle FXC using their lengths $\cos\angle FXC = \frac{'150'+'50'-'300'}{2\times\sqrt{'150'}\times\sqrt{'50'}} \Rightarrow \angle FXC = \left(125.2643...^{\circ}\right)$ NB: Check that the cosine rule they are using is to find the required angle.
	A1	For the correct angle with awrt. $\angle FXC = 125^{\circ}$ Accept answers which round to 125°, but they must round to this value.
	ALT – Uses the right-angled triangle *FGX*	
	M1	Let X be the midpoint of BH. For any method (Pythagoras or trigonometry) to find the length GX $GX = \frac{\sqrt{200}}{2} = \left(\sqrt{50} \text{ or } 5\sqrt{2} \text{ or awrt } 7.07\right)$ If they use the sin ratio to find the angle, identify $FG = 10$ (cm)
	M1	For any method (Pythagoras or trigonometry) to find the length FX $FX = \sqrt{50+100} = \left(\sqrt{150} \text{ or } 5\sqrt{6} \text{ awrt } 12.2\right)$ or $FX = \sqrt{200-50} = \left(\sqrt{150} \text{ or } 5\sqrt{6} \text{ awrt } 12.2\right)$ **OR** If they use the tan ratio to find the angle, identify $FG = 10$ (cm)
	A1	For both correct lengths $\sqrt{50}$ and 10 for tan **or** $\sqrt{50}$ and $\sqrt{150}$ for cos **or** $\sqrt{150}$ and 10 for sin Allow decimal equivalents, i.e., 7.07 and 12.2
	M1	$\cos\angle FXG = \frac{\sqrt{50}}{\sqrt{150}} \Rightarrow \angle FXG = \left(54.7356...^{\circ}\right)$, $\sin\angle FXG = \frac{10}{\sqrt{150}} \Rightarrow \angle FXG = \left(54.7356...^{\circ}\right)$, $\tan\angle FXG = \frac{10}{\sqrt{50}} \Rightarrow \angle FXG = \left(54.7356...^{\circ}\right)$ So the required angle is $\angle FXC = 180 - '54.7356...'^{\circ} = \left(125.264...^{\circ}\right)$ We must see this calculation for the award of this mark. Do **not** award just for 54.7356…°
	A1	For the correct angle with awrt. $\angle FXC = 125^{\circ}$ Accept answers which round to 125°, but they must round to this value.

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Question	Scheme	Mark

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3(a)		
	$AM = \sqrt{10^2 - 8^2}$, $= 6$	M1,A1 (2)
(b)		
	$\cos C = \frac{26^2+16^2-26^2}{2\times16\times26} = \frac{256}{832}$ $\left(=\frac{4}{13}\right.$ oe$\left.\right)$	M1A1
	$\angle BCD = 72°$	A1cao (3)
(c)	$AD = \sqrt{26^2 - 10^2} = 24$ or $DM = \sqrt{26^2 - 8^2} = 6\sqrt{17}$ oe (24.73...)	M1A1
	$\tan(\angle DMA) = \frac{24}{6}$ or $\cos(\angle DMA) = \frac{6}{6\sqrt{17}}$ or $\sin(\angle DMA) = \frac{24}{6\sqrt{17}}$	M1
	$\angle DMA = 76°$	A1cao (4) **[9]**

(a)	
M1	Use Pythagoras, with a minus sign, to obtain the length of AM.
A1	Correct length obtained.
NB	Answers w/o working get both marks (use of (3,4,5) triangle)
(b)	
M1	Use the cosine rule in ΔBCD to obtain a numerical expression for cos C. Correct formula in either form may be quoted and substitution attempted or correct formula can be implied by the correct substitution. Complete method needed, so if another angle found first do not award this mark until a value for angle BCD is obtained.
A1	Correct value for the cosine obtained (Decimal to be awrt 0.308 (0.30769...)) Award by implication if final answer is awrt 72°.
A1cao	For 72° (from correct working)
ALT	**Use the isosceles triangle**
M1A1	$\cos C = \frac{8}{26}$ oe (Any trig function allowed provided work completed to a value for angle BCD)
A1cao	For 72° (from correct working)
(c)	
M1	Attempt the length of AD or DM using Pythagoras with a minus sign.
A1	Correct value for their choice of line,
M1	Use an appropriate trig function. The length of AD or DM must have been attempted with a + or a - sign.
A1cao	Correct answer.
	Penalise once only in (b) and (c) for failing to round as instructed.

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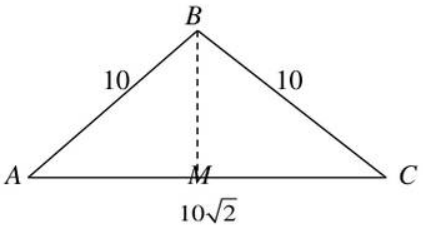
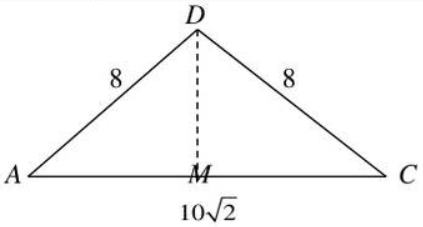
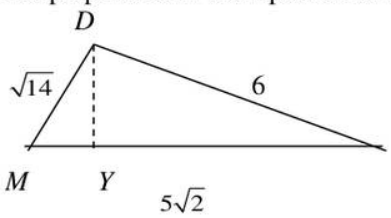
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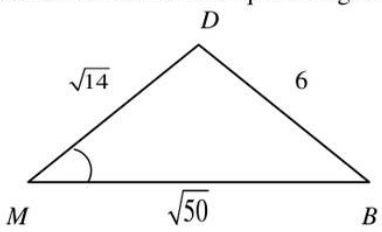
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9 (a)	$AC = \sqrt{10^2 + 10^2} = \sqrt{200} = 10\sqrt{2}$	M1A1 [2]
(b)	 $BM = \sqrt{10^2 - \left(5\sqrt{2}\right)^2} = 5\sqrt{2}$	M1A1 [2]
(c)	 $DM = \sqrt{8^2 - \left(5\sqrt{2}\right)^2} = \sqrt{14}$ $\angle DMB = \cos^{-1}\left(\frac{14 + 50 - 36}{2 \times \sqrt{14} \times 5\sqrt{2}}\right) = 58.0519...^\circ \approx 58.1^\circ$	M1A1 M1A1 [4]
(d)	Let perpendicular from point D to BM be the point Y  Vertical height of shape $ABCD$ In triangle DMY (above) Height $= DY = \sqrt{14}\sin 58.0519^\circ = 3.1749... \approx 3.17$ (cm)	M1A1 [2]
		Total 10 marks

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Part	Mark	Notes
(a)	M1	For using Pythagoras theorem correctly on triangle ABC to find AC
	A1	For $AC = 10\sqrt{2}$
(b)	M1	For using Pythagoras theorem or any appropriate trigonometry correctly on triangle ABC to find BM
	A1	For the correct length $BM = 5\sqrt{2}$ **NB:** Allow for the answer just seen without any working as the triangle ABC is isosceles.
(c)	M1	For using Pythagoras theorem to find the length DM
	A1	For $DM = \sqrt{14}$ or accept awrt 3.74
	M1	For using cosine rule to find the required angle DMB 
	A1	For awrt 58.1°
(d)	M1	For using any appropriate trigonometry to find the required length. Ft their angle DMB
	A1	For awrt 3.17 (cm) Accept 3.18 (cm)

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 9 of 22 · Question 9 part a · 2 marks

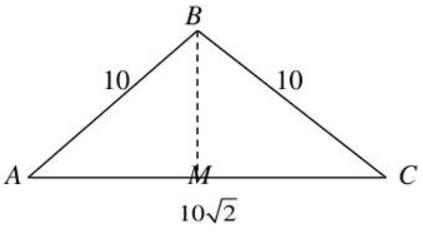
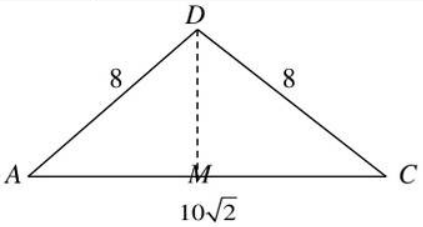
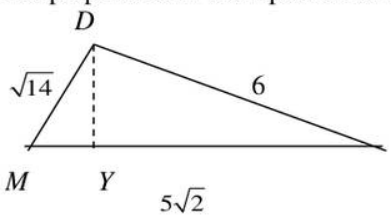
2021 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 10 of 22 · Question 9 part b · 2 marks

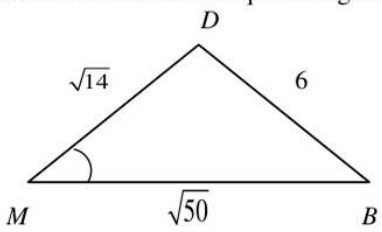
2021 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

9 (a)	$AC = \sqrt{10^2 + 10^2} = \sqrt{200} = 10\sqrt{2}$	M1A1 [2]
(b)	 $BM = \sqrt{10^2 - \left(5\sqrt{2}\right)^2} = 5\sqrt{2}$	M1A1 [2]
(c)	 $DM = \sqrt{8^2 - \left(5\sqrt{2}\right)^2} = \sqrt{14}$ $\angle DMB = \cos^{-1}\left(\frac{14 + 50 - 36}{2 \times \sqrt{14} \times 5\sqrt{2}}\right) = 58.0519...^\circ \approx 58.1^\circ$	M1A1 M1A1 [4]
(d)	Let perpendicular from point D to BM be the point Y  Vertical height of shape $ABCD$ In triangle DMY (above) Height $= DY = \sqrt{14}\sin 58.0519^\circ = 3.1749... \approx 3.17$ (cm)	M1A1 [2]
		Total 10 marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 10 of 22 · Question 9 part b · 2 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For using Pythagoras theorem correctly on triangle ABC to find AC
	A1	For $AC = 10\sqrt{2}$
(b)	M1	For using Pythagoras theorem or any appropriate trigonometry correctly on triangle ABC to find BM
	A1	For the correct length $BM = 5\sqrt{2}$ **NB:** Allow for the answer just seen without any working as the triangle ABC is isosceles.
(c)	M1	For using Pythagoras theorem to find the length DM
	A1	For $DM = \sqrt{14}$ or accept awrt 3.74
	M1	For using cosine rule to find the required angle DMB 
	A1	For awrt 58.1°
(d)	M1	For using any appropriate trigonometry to find the required length. Ft their angle DMB
	A1	For awrt 3.17 (cm) Accept 3.18 (cm)

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 10 of 22 · Question 9 part b · 2 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 11 of 22 · Question 9 part a · 2 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

9(a)	$AC = \sqrt{12^2 + 12^2} = \sqrt{288} = 12\sqrt{2}$	M1A1 [2]
(b)	$x = \dfrac{\dfrac{12\sqrt{2}}{2}}{\cos 30^\circ} = \sqrt{96} = \left(4\sqrt{6}\right)^*$	M1A1 cso [2]
(c)	$\cos \angle AOB = \dfrac{96 + 96 - 12^2}{2 \times \sqrt{96} \times \sqrt{96}} \Rightarrow \angle AOB = 75.522...^\circ$ Area $\Delta OAB = \dfrac{1}{2} \times \sqrt{96} \times \sqrt{96} \times \sin' 75.522^\circ ' = \left(46.4758...\right)$ $4 \times 46.4758 + 12^2 = 329.90.... \Rightarrow$ Area = 330 [m^2]	M1A1 M1 M1A1 [5]
(d)	$\left[\angle OBC = \dfrac{180 - 75.522}{2} = 52.239^\circ\right]$ Let Y on OB be the foot of the perpendicular from A to OB $AY = 12 \sin 52.239^\circ = \left(9.4868...\right)$ $\cos \angle AYC = \dfrac{'9.4868'^2 + '9.4868'^2 - 288}{2 \times '9.4868' \times '9.4868'} \Rightarrow \angle AYC = 126.87... \approx 127^{[o]}$	M1 A1 M1A1 [4]
		Total 13 marks

Question	Notes	Marks
9(a)	For using Pythagoras theorem on triangle ABC or triangle ADC $AC = \sqrt{12^2 + 12^2} = ...$	M1
	For the correct value of AC $AC = \sqrt{288} = 12\sqrt{2}$	A1 [2]
(b)	Let the intersection of AC and BD be X Uses any appropriate trigonometry on triangle OAX For example; $x = \dfrac{\dfrac{12\sqrt{2}}{2}}{\cos 30^\circ} = ...$ **OR** $\dfrac{\sin 120^\circ}{12\sqrt{2}} = \dfrac{\sin 30^\circ}{OA} \Rightarrow OA = ...$ OR $\left(12\sqrt{2}\right)^2 = x^2 + x^2 - 2 \times x \times x \cos 120 \Rightarrow x = ...$	M1
	For the correct value of x	A1 cso

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 11 of 22 · Question 9 part a · 2 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

	$x = \sqrt{96} = \left(4\sqrt{6}\right)$ *	[2]
(c)	For using trigonometry to find angle $\angle AOB$ $\cos \angle AOB = \frac{96+96-12^2}{2\times\sqrt{96}\times\sqrt{96}} = ...$	M1
	$\angle AOB = 75.522...^{\circ}$	A1
	Area of triangle OAB Area $= \frac{1}{2}\times\sqrt{96}\times\sqrt{96}\times\sin'75.522^{\circ}' = \left(46.4758...\right)$	M1
	Total area of the pyramid $= 4\times46.4758+12^2 = \left(329.90....\right)$	M1
	For the correct final area = awrt 330 (cm^2)	A1 [5]
	ALT	
	Height of the triangle of one of the triangular faces: $h = \sqrt{\left(4\sqrt{6}\right)^2 - 6^2} = ...$	M1
	$h = 2\sqrt{15}$	A1
	Area of triangle $OAB = \frac{1}{2}\times2\sqrt{15}\times12 = 12\sqrt{15} = \left(46.4758...\right)$	M1
	Total area of the pyramid $= 4\times12\sqrt{15}+144 = \left(329.903...\right)$	M1
	For the correct final area = awrt 330 (cm^2) Allow an exact answer of $48\sqrt{15}+144$ $(cm)^2$ oe	A1
(d)	$\cos \angle AOB = \frac{96+96-12^2}{2\times\sqrt{96}\times\sqrt{96}} = 75.522^{\circ}$ or $\angle OBC = \frac{180^{\circ}-75.522^{\circ}}{2} = 52.239^{\circ}$	
	Let Y on OB be the foot of the perpendicular from A to OB Length $AY = 12\sin 52.239^{\circ} = \left(9.4868...\right)$ OR Length $AY = 4\sqrt{6}\sin 75.522^{\circ} = \left(9.4868...\right)$	M1A1 [M1A1]
	For the appropriate trigonometry on triangle AYC to find angle AYC $\cos \angle AYC = \frac{'9.4868'^2+'9.4868'^2-288}{2\times'9.4868'\times'9.4868'} \Rightarrow \left(\angle AYC = 126.87\right)$	M1
	Angle between plane AOB and plane OBC =awrt 127[°]	A1 [4]
		Total 13 marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 11 of 22 · Question 9 part a · 2 marks

2023 JAN · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 12 of 22 · Question 6 part a · 3 marks

2023 NOV · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

6(a)	$(AC=)\sqrt{x^2+x^2}\ (=\sqrt{2}x)$ **or** $(AC=)\dfrac{x}{\sin 45}$ **or** $(AC=)\dfrac{x}{\cos 45}$ $(AO=)\sqrt{\left(\dfrac{x}{2}\right)^2+\left(\dfrac{x}{2}\right)^2}\ \left(=\sqrt{\dfrac{x^2}{2}}=\dfrac{x}{\sqrt{2}}\right)$ **or** $(AO=)\dfrac{\frac{x}{2}}{\sin 45}$ **or** $(AO=)\dfrac{\frac{x}{2}}{\cos 45}$	M1
	$(VO=)\sqrt{x^2-\left(\dfrac{"\sqrt{2}x"}{2}\right)^2}$ **or** $\sqrt{x^2-\left(\dfrac{1}{2}\sqrt{x^2+x^2}\right)^2}$ **or** $\sqrt{x^2-\dfrac{x^2}{2}}\ \left(=\sqrt{\dfrac{1}{2}x^2}\right)$ **or** $(VO=)\dfrac{\sqrt{2}}{2}\tan 45$	M1
	$=\dfrac{\sqrt{2}}{2}x$ (cm)*	A1*cso [3]
ALT	Any indication the candidate realises triangle AVC is right angled and isosceles.	M1
	$\sin 45=\dfrac{VO}{x}$ **or** $x\sin 45=VO$	M1
	$VO=\dfrac{\sqrt{2}}{2}x$ *	A1*cso [3]
(b)	$(\angle AVC=)\cos^{-1}\left(\dfrac{x^2+x^2-\left(\sqrt{2}x\right)^2}{2\times x\times x}\right)$ **or** $(\angle AVC=)2\times\sin^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{x}\right)$ **or** $(\angle AVC=)2\times\cos^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{x}\right)$ **or** $(\angle AVC=)2\times\tan^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{\frac{\sqrt{2}}{2}x}\right)$	M1
	$=90^\circ$	A1 [2]

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 12 of 22 · Question 6 part a · 3 marks

2023 NOV · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

(c)	Let M be the midpoint of AB and let N be the midpoint of DC	
	$(VM = VN =)\sqrt{x^2 - \left(\frac{x}{2}\right)^2} = \left(\frac{\sqrt{3}}{2}x\right)$	M1
	$\cos MVN = \dfrac{\left("\frac{\sqrt{3}}{2}x"\right)^2 + \left("\frac{\sqrt{3}}{2}x"\right)^2 - x^2}{2\times\left("\frac{\sqrt{3}}{2}x"\right)\times\left("\frac{\sqrt{3}}{2}x"\right)}$ **or**	
	$(\angle MVN =)\cos^{-1}\left(\dfrac{\left("\frac{\sqrt{3}}{2}x"\right)^2 + \left("\frac{\sqrt{3}}{2}x"\right)^2 - x^2}{2\times\left("\frac{\sqrt{3}}{2}x"\right)\times\left("\frac{\sqrt{3}}{2}x"\right)}\right)$ **or**	M1
	$(\angle MVN =)2\times\cos^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{"\frac{\sqrt{3}}{2}x"}\right)$ **or** $(\angle MVN =)2\times\sin^{-1}\left(\dfrac{\frac{1}{2}x}{"\frac{\sqrt{3}}{2}x"}\right)$	
	or $(\angle MVN =)2\times(\angle MVO =)$ **and** $\cos MVO = \dfrac{\frac{\sqrt{2}}{2}x}{"\frac{\sqrt{3}}{2}x"}$	
	$70.528...^\circ \approx 70.5^\circ$	A1 [3]
	ALT	M1
	$OM = \frac{x}{2}$	
	$(\angle MVN =)2\tan^{-1}\left[\dfrac{\frac{x}{2}}{\frac{\sqrt{2}x}{2}}\right] = \left(2\tan^{-1}\left[\frac{1}{\sqrt{2}}\right] =\right)70.528...^\circ \approx 70.5^\circ$	M1A1 [3]
(d)	Volume $= \frac{1}{3}\times\left(\frac{\sqrt{2}}{2}x\right)\times x^2\left(=\frac{\sqrt{2}}{6}x^3\right)$ oe	M1
	$200 = \frac{1}{3}\times\left(\frac{\sqrt{2}}{2}x\right)\times x^2 \Rightarrow x =$	dM1
	$x = 9.4672... \approx 9.47(\text{cm})$	A1 [3]
		Total 11 marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 12 of 22 · Question 6 part a · 3 marks

2023 NOV · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

Part	Mark	Notes
For part (a), we must see the algebraic proof, including x throughout.		
(a)	M1	For correct use of Pythagoras theorem or any correct appropriate trigonometry and a complete method to find the length AC or AO ie candidates must rearrange correctly – minimum steps as shown in mark scheme and no errors must be present.
	M1	For a full, correct and complete method to find VO, using their AC/AO.
	A1 cso	For $\frac{\sqrt{2}}{2}x$ (cm) – correct solution only, no errors or omissions, minimum steps as shown.
ALT	M1	Any indication the candidate realises triangle AVC is right angled and isosceles. This mark may be implied by correct working for the next method mark or present on the diagram.
	M1	For the correct trig minimum steps as shown and no errors.
	A1 cso	For $\frac{\sqrt{2}}{2}x$ (cm) – correct solution only, no errors or omissions.
(b)	M1	For any correct appropriate trigonometry to find angle AVC. $(\angle AVC =)$ If finding $\angle AVO$, must be doubled for this mark. AC must be $\sqrt{2}x$ oe, AO must be $\frac{\sqrt{2}}{2}x$ oe.
	A1	For 90°
	Allow M1A1 for candidates writing down the correct answer by inference.	
Consistent omission of x is permissible for all marks in part (b).		
(c)	M1	For correct Pythagoras theorem to find the lengths of VM or VN
	M1	For any correct trigonometry to find angle MVN, such as the statements shown in the mark scheme. The candidate can state, for example, cos(MVN) = fully correct expression (using their VN and VM) or, for example, $\cos^{-1}$(fully correct expression), where they do not have to write angle MVN = i.e – this can involve any fully correct method to reach an equation which would allow them to use inverse trig to find MVN or if they use inverse trig, a fully correct expression must be seen with the correct inverse trig function. Send any items where allocation of marks is unclear (potentially using other methods to the main and ALT) to Review. or fully correct **method** to find angle MVO which must then be doubled for this mark.
	A1	For any answer stated which rounds to 70.5° (ie more decimal places is condoned).
Consistent omission of x is permissible for all marks in part (c).		
ALT	M1	States or uses $OM = \frac{x}{2}$
	M1	For fully correct **method** to find angle MVO which must then be doubled for this mark.
	A1	For any answer stated which rounds to 70.5° (ie more decimal places is condoned).
(d)	M1	For a correct unsimplified expression for the volume of the pyramid as shown.
	dM1	For placing their expression = 200, which must involve a term in x^3, and a fully correct method/rearrangement to solve. Note this mark is dependent on the first method mark, so they must have got the correct unsimplified expression even if then simplified incorrectly.
	A1	For any final answer stated which rounds to 9.47 (ie more decimal places is condoned).

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 12 of 22 · Question 6 part a · 3 marks

2023 NOV · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 13 of 22 · Question 8 part a · 2 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

8(a)	$\tan 30° = \frac{VP}{6}$ where P is the centre of the base $VP = 6\tan 30° = 2\sqrt{3}$ *	M1 A1*cso [2]
(b)	$AC = \sqrt{12^2 + 12^2} = 12\sqrt{2}$ or $AC = \frac{12}{\sin 45}$ or $AC = \frac{12}{\cos 45}$ $VA^2 = \left(2\sqrt{3}\right)^2 + \left(6\sqrt{2}\right)^2 (= 84)$ $VA = 2\sqrt{21}$ (cm) oee	M1 oe M1 oe A1 [3]
(c)	$VX = \sqrt{\left('2\sqrt{21}'\right)^2 - 6^2} \left(= 4\sqrt{3}\right)$ or $VX = \frac{6}{\cos 30}$ or $VX = \sqrt{6^2 + \left(2\sqrt{3}\right)^2}$ where X is the midpoint of the side AD Area $= \frac{1}{2} \times 4\sqrt{3} \times 12 = 24\sqrt{3}$ (cm^2) oee	M1 M1A1 [3]
(d)	$'24\sqrt{3}' = \frac{1}{2} \times '2\sqrt{21}' \times DY$ where Y is the foot of the perpendicular from D to VA $DY = \frac{24\sqrt{7}}{7}$ oee	M1 A1 [2]
(e)	$\cos\theta = \frac{\left(,\frac{24\sqrt{7}}{7},\right)^2 + \left(,\frac{24\sqrt{7}}{7},\right)^2 - \left('12\sqrt{2}'\right)^2}{2 \times ,\frac{24\sqrt{7}}{7}, \times ,\frac{24\sqrt{7}}{7},} = -\frac{3}{4}$ $\theta = 138.6°$	M1A1 A1 [3]
		Total is 13 marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 13 of 22 · Question 8 part a · 2 marks

2025 JUN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For forming a correct equation using trigonometry. e.g. $\tan 60° = \frac{6}{VP}$ or $\frac{h}{\sin 30°} = \frac{6}{\sin 60°}$ (If attempted by ratio, send to review)
	A1*cso	For **rearranging** to show the required result. Minimum need to see one intermediate step from their equation to the answer ,if the height is not the subject of their equation: e.g. $\tan 30° = \frac{VP}{6} \Rightarrow VP = 6\tan 30° = 2\sqrt{3}$ is M1A1 $VP = 6\tan 30° = 2\sqrt{3}$ is M1A1 $\tan 30° = \frac{VP}{6} \Rightarrow \frac{\sqrt{3}}{3} = \frac{VP}{6} \Rightarrow VP = 2\sqrt{3}$ is M1A1 $\tan 30° = \frac{VP}{6} \Rightarrow VP = 2\sqrt{3}$ is M1A0
(b)	M1oe	Correct use of Pythagoras' theorem or trigonometry to obtain the length/half length of the diagonal of the base AC or $\frac{1}{2}AC$ Or other valid method to find VM, (Let M to be the midpoint of AB) e.g.: $VM = \frac{6}{\cos 30}\left(= 4\sqrt{3}\right)$ or $VM = \sqrt{6^2 + \left(2\sqrt{3}\right)^2}\ \left(= 4\sqrt{3}\right)$
	M1oe	Use Pythagoras' theorem to obtain a numerical value for VA / VA^2 Any correct method. e.g. $VA^2 = \left(4\sqrt{3}\right)^2 + (6)^2$
	A1	Correct exact length $2\sqrt{21}$ or $\sqrt{84}$ oee (or exact equivalent)
(c)	M1	Correct use of Pythagoras' theorem or trigonometry to obtain the length of VX, which is the same as VM, could be found in part b Or correct use of cosine rule (ft their VA) to find the value of cosine value of any angle in triangle VAD e.g. $AD^2 = VA^2 + VA^2 - 2VA \times VA \times \cos \angle AVD$ $12^2 = \left('2\sqrt{21}'\right)^2 + \left('2\sqrt{21}'\right)^2 - 2 \times '2\sqrt{21}' \times '2\sqrt{21}' \times \cos \angle AVD$ $\left\{\Rightarrow \cos \angle AVD = \frac{1}{7}\right\}$
	M1	Correct method to find the area of triangle VAD (Do not have to use exact values) Accept any correct valid method. e.g. $\left\{\cos \angle AVD = \frac{1}{7} \Rightarrow \sin \angle AVD = \frac{4\sqrt{3}}{7}\right\} \Rightarrow A = \frac{1}{2} \times '2\sqrt{21}' \times '2\sqrt{21}' \times '\frac{4\sqrt{3}}{7}' = 24\sqrt{3}$
	A1	Correct **exact** value for area
(d)	M1	Use of area of triangle with VA as the base to form an equation for DY Or other valid method: e.g. using trigonometry ratio: $\sin \angle VAD = \frac{'4\sqrt{3}'}{'2\sqrt{21}'} = \frac{DY}{12}$

Type MS · Lengths in a 3D solid using right-angled triangles

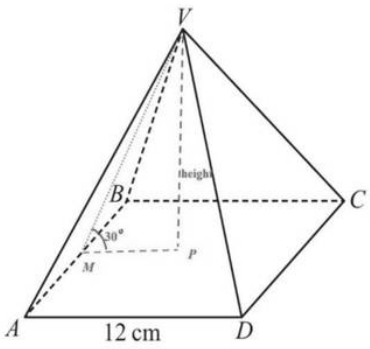
Answer 13 of 22 · Question 8 part a · 2 marks

2025 JUN · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

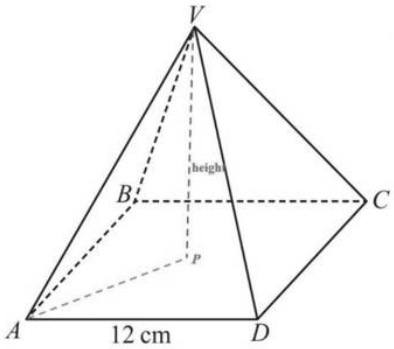
	A1	Correct exact length $DY = \frac{24\sqrt{7}}{7}$ oee, e.g. $DY = \frac{8\sqrt{63}}{7}$ or $24\sqrt{\frac{1}{7}}$
(e)	M1	Correct use of cosine rule: $\cos\theta = \frac{"DY"^2 + BY"^2 - "BD"^2}{2\times"DY"\times"BY"}$ with their values Or other valid method, e.g. $\sin\frac{\theta}{2} = \frac{"\frac{BD}{2}"}{"DY"} = \frac{"6\sqrt{2}"}{"\frac{24\sqrt{7}}{7}"}$
	A1	Obtains $\cos\theta = -\frac{3}{4}$ or $\sin\frac{\theta}{2} = \frac{\sqrt{14}}{4}$
	A1	Correct angle found

Useful diagrams:

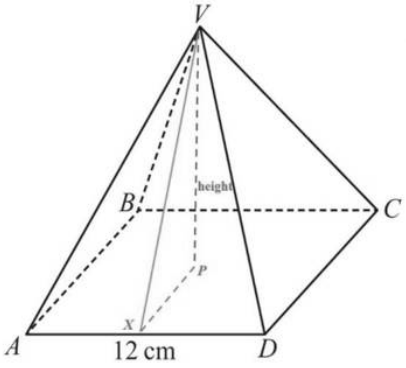
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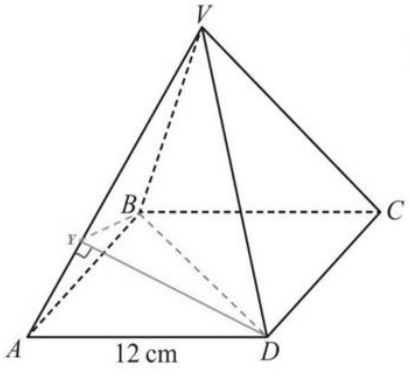
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4



Type MS · Lengths in a 3D solid using right-angled triangles

Answer 13 of 22 · Question 8 part a · 2 marks

2025 JUN · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 14 of 22 · Question 8 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

8 (a)	$BN = 10\cos 60^\circ$ $= 5$ (cm)*	M1 A1cso [2]
(b)	$FN^2 = 10^2 - 5^2 = 75$ or $FN = 10\sin 60 = 5\sqrt{3}$ or $FN = 5\tan 60$ $NE = \sqrt{12^2 + 75} = 14.7986... \approx 14.8$ (cm)	M1 M1A1 [3]
(c)	Let R be the midpoint of FN and P be the point on AD that corresponds to point N on BC. $\angle RPN = \tan^{-1}\left(\dfrac{\sqrt{75}/2}{12}\right) = 19.84167...^\circ \approx 19.8^\circ$	M1A1 [2]
(d)	Let the perpendicular from Y to line BC meet BC at Z $AY = \sqrt{AZ^2 + ZY^2} = \sqrt{12^2 + 6.5^2 + \left(\dfrac{\sqrt{75}}{2}\right)^2} = \sqrt{205}$ $FC = \sqrt{FN^2 + NC^2} = \sqrt{75 + 3^2} = \sqrt{84} \Rightarrow FY = \sqrt{21}$ So $EY = \sqrt{12^2 + 21} = \sqrt{165}$ $\angle AYE = \cos^{-1}\left(\dfrac{205 + 165 - 100}{2 \times \sqrt{205} \times \sqrt{165}}\right) = 42.7745...^\circ \approx 42.8^\circ$	M1A1 M1 M1 M1A1 [6]
		Total 13 marks

(a)	M1	Use cosine or sine in triangle BNF, or any complete method. $BN = 10\cos 60^\circ$ or $\cos 60^\circ = \dfrac{BN}{10}$ or $\dfrac{BN}{\sin 30} = \dfrac{10}{\sin 90}$ or $BN = 10\sin 30$ or $FN = 10\sin 60 = 5\sqrt{3}$ and $BN^2 = 10^2 - (5\sqrt{3})^2$ (may use any single letter instead of BN)
	A1 cso	$BN = 5$ (cm) Correct answer from correct working. (Answer given.) Must show a calculation in the form $BN = \ldots$ e.g. $BN = 10\cos 60$
(b)	M1	Use Pythag or trig to find a numerical expression for FN or FN^2 eg $FN^2 = 10^2 - 5^2$ $(= 75)$ or $FN = 10\sin 60$ $(= 5\sqrt{3} = 8.66...)$ or $FN = 5\tan 60$
	M1	Use Pythag with their FN to find a numerical expression for NE, eg $NE = \sqrt{12^2 + (5\sqrt{3})^2}$ $(= \sqrt{219})$
	A1	$NE = 14.8$ (cm) **given to 3SF**.
(c)	M1	Use tangent with their FN, eg $\tan RPN = \left(\dfrac{\frac{1}{2}\sqrt{75}}{12}\right)$ or any complete method.
	A1	19.8° **given to 1DP**.

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 14 of 22 · Question 8 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

(d)	M1	Complete method to find AY eg $AY=\sqrt{(12^2+6.5^2)+\left(\frac{\sqrt{75}}{2}\right)^2}$ $(=\sqrt{205}=14.31...)$ (allow their FN)
	A1	$AY=\sqrt{205}$ $(=14.31\ldots)$ (may be implied by subsequent working)
	M1	Complete method to find FY eg $FC=\sqrt{75+3^2}\left(=\sqrt{84}\right)\Rightarrow FY=\frac{1}{2}\sqrt{84}$ $(=\sqrt{21})$ (allow their FN) eg $FC=\sqrt{8^2+10^2-2\times8\times10\cos60}\left(=\sqrt{84}\right)\Rightarrow FY=\frac{1}{2}\sqrt{84}$ $(=\sqrt{21}=4.58...)$
	M1	Method to find EY using their FY (correct or from a correct method) eg $EY=\sqrt{12^2+21}$ $(=\sqrt{165}=12.84...)$
	M1	Substitute their values for AY and EY into the cosine rule used in triangle AYE. $\cos AYE=\left(\frac{205+165-100}{2\times\sqrt{205}\times\sqrt{165}}\right)$ oe
	A1	$\angle AYE=42.8^\circ$ **given to 1DP**. Allow more than 1DP if 1DP was penalised in part (c).

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 14 of 22 · Question 8 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 15 of 22 · Question 9 part a · 1 mark

2022 JAN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

9 (a)	$\cos\alpha = \frac{3}{\sqrt{13}}$	B1 (1)
(b)	$h = \sqrt{17^2 - \left(9^2 + 12^2\right)} = 8$	M1 M1 A1 (3)
(c)	Let M be the midpoint of BC $\tan\theta^\circ = \frac{h}{OM} = \frac{8}{12} = \frac{2}{3}$ *	M1 A1 (2)
(d)	Let N be the midpoint of EM	
	$EM = \sqrt{8^2 + 12^2} = 4\sqrt{13}$	M1
	$NO = \sqrt{12^2 + \left(2\sqrt{13}\right)^2 - 2(12)\left(2\sqrt{13}\right)\left(\frac{3}{\sqrt{13}}\right)} \Rightarrow NO = 2\sqrt{13}$	M1
	Hence triangle ONM is isosceles	
	$180 - 2\times 33.7 = 112.6^\circ$	M1 A1 (4)
		Total 10 marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 15 of 22 · Question 9 part a · 1 mark

2022 JAN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 16 of 22 · Question 11 part a · 4 marks

2019 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

11(a)	$AC = \sqrt{(16^2 + 16^2)} = 16\sqrt{2}$	M1A1
ALT:	$AP^2 + PD^2 = 16^2 \Rightarrow AP = 8\sqrt{2}$	
	$VP = 8\sqrt{2}\tan 45 = 8\sqrt{2}$ (where P is the centre of the base) *	M1A1cso (4)
(b)	$VA^2 = \left(8\sqrt{2}\right)^2 + \left(8\sqrt{2}\right)^2 (= 256)$ or $VA = \dfrac{8\sqrt{2}}{\sin 45°}$	M1A1
	$VA = 16$ cm	A1 (3)
(c)	$DX^2 = 16^2 - 8^2$ where X is the foot of the perpendicular from D to VA	M1A1
	$DX = 8\sqrt{3}$	A1 (3)
(d)	$\tan\theta = \dfrac{8\sqrt{2}}{8}, \sin\theta = \dfrac{8\sqrt{2}}{8\sqrt{3}}, \cos\theta = \dfrac{8}{8\sqrt{3}}$	M1A1
	(or unsimplified if cosine or sine rule used) $\theta = 54.7°$	A1 (3)
(e)	$\cos\phi = \dfrac{\left(8\sqrt{3}\right)^2 + \left(8\sqrt{3}\right)^2 - \left(16\sqrt{2}\right)^2}{2\times 8\sqrt{3}\times 8\sqrt{3}} \left(= -\dfrac{1}{3}\right)$	M1A1
	$\phi = 109.5°$	A1 (3) [16]

(a)	
M1	Use Pythagoras (with a + sign) to obtain the length of the diagonal of the base. Or use Pythagoras with correct sign to obtain the half diagonal
A1	Correct length for the diagonal or half diagonal
M1	Use tan in ΔAPV, their AP and angle of 45° to obtain the height
A1cso	Correct answer with no errors in the working
	OR: State ΔAVP is isosceles, or shown the 2 correct angles – can be on a diagram M1
	$VP(= AP) = 8\sqrt{2}$ A1
	OR use any other complete valid method M1 Correct result A1
(b)	
M1	Use Pythagoras or trigonometry in ΔAPV or ΔAVC (or any other complete, valid method) to obtain a numerical expression for VA.
A1	Correct numbers in their choice of method.
A1	Correct length obtained.
	$AV = 16$ w/o working scores M1A1A1

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 16 of 22 · Question 11 part a · 4 marks

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Question Number	Scheme	Marks
(c)		
M1	Use Pythagoras with a minus sign (seen or implied) or trigonometry in ΔADX OR any other complete valid method **NB**: triangle ΔADV is equilateral.	
A1	Correct numbers in their choice of method.	
A1	Correct **exact** length for the perpendicular.	
(d)		
M1	Identify the correct triangle needed with the required angle marked (may be on Figure 1). This may be shown explicitly or implied by their work that follows.	
A1	Reach one of $\tan\theta = \frac{8\sqrt{2}}{8}$, $\sin\theta = \frac{8\sqrt{2}}{8\sqrt{3}}$, $\cos\theta = \frac{8}{8\sqrt{3}}$ oe	
A1	Correct answer, **must be 1 dp.**	
(e)		
M1	Use cosine rule: $\cos\phi = \frac{"DX"^2 + "XB"^2 - "BD"^2}{2\times"DX"\times"XB"}$ (their values)	
A1	Correct numbers substituted, follow through their previous answers	
A1	Correct answer, **must be 1 dp unless already penalised in (d)**	
	Any other routes should be marked: M1 Correct, complete method (ie it must be possible to reach a value for the required angle) A1 Correct numbers substituted A1 Correct answer, **must be 1 dp unless already penalised in (d)**	

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 16 of 22 · Question 11 part a · 4 marks

2019 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

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Type MS · Lengths in a 3D solid using right-angled triangles

Answer 17 of 22 · Question 5 part a · 3 marks

2023 JAN · Paper 02 · find part a in this mark scheme extract

5 a	$OC = \frac{1}{2}\sqrt{12^2+12^2} = \left[6\sqrt{2}\right]$ or $AC = \sqrt{12^2+12^2} = \left[12\sqrt{2}\right]$	M1
	$h = EO = 6\sqrt{2} \times \frac{\sqrt{3}}{3} = 2\sqrt{6}$ oe	M1 A1
		(3)
b	Midpoint of AD to $F = \left(6 - \frac{12}{1+4}\right)$ [= 3.6]	M1
	$OF = \sqrt{6^2+3.6^2} = \frac{6\sqrt{34}}{5}$	M1
	$\tan\theta = \frac{2\sqrt{6}}{\frac{6\sqrt{34}}{5}} = 35^\circ$	M1 A1
		(4)
		Total 7 marks

USEFUL SKETCH

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 18 of 22 · Question 10 part a · 3 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

10(a)	$AC = \sqrt{(2x)^2 + x^2} = (\sqrt{5}x)$ $VA = '\sqrt{5}x' \sin 45^o = \frac{\sqrt{10}}{2}x$ *	B1 M1A1 cso [3]
(b)	Height $= \sqrt{\left(\frac{\sqrt{10}}{2}x\right)^2 - \left(\frac{\sqrt{5}}{2}x\right)^2} = \frac{\sqrt{5}}{2}x$	M1A1 [2]
(c)	$\cos \angle VBA = \frac{1}{\frac{\sqrt{10}}{2}}$ $\Rightarrow \angle VBA = 50.768...^o \approx 50.8^o$ awrt	M1 A1 [2]
(d)	Let M be the point at which the diagonals AC and BD meet. The required angle is either AMB or DMC AM = MB $= \frac{AC}{2} = \frac{\sqrt{5}x}{2}$ or DM=MC=$\frac{AC}{2} = \frac{\sqrt{5}x}{2}$ $\cos \angle AMB = \frac{\left(\frac{\sqrt{5}}{2}\right)^2 + \left(\frac{\sqrt{5}}{2}\right)^2 - 2^2}{2 \times \frac{\sqrt{5}}{2} \times \frac{\sqrt{5}}{2}}$ oe $(\cos \angle AMB = -\frac{3}{5} \Rightarrow \angle AMB = 126.869...^o) \approx 126.9^o$ awrt	B1 B1 M1 A1 [4]
(e)	$9\sqrt{5} = \frac{1}{3} \times 2x \times x \times "\frac{\sqrt{5}}{2}x" \Rightarrow x = \cdots$ $x = 3$	M1 A1 [2]
		Total 13 marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 18 of 22 · Question 10 part a · 3 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For using correct Pythagoras theorem to find length *AC or ½ AC*, explicit substitution must be seen.
	M1	For correct trigonometry using their AC in triangle VAC or half of their AC in triangle VAM or correct Pythagoras theorem (let M be the centre of the base at which the diagonals *AC* and *BD* meet) to find length *VA*. *e.g.* $VA = \sqrt{5}x\cos(45°) = \frac{\sqrt{10}}{2}x$ *or* $VA = \frac{\sqrt{5}}{2}x \div \cos(45°) = \frac{\sqrt{10}}{2}x$ *or* $VA = \frac{\sqrt{5}}{2}x \div \sin(45°) = \frac{\sqrt{10}}{2}x$
	A1 cso	Achieves printed answer with no errors. Condone AV instead of VA Note: $\frac{\sqrt{10}}{2}x$ *written as* $\frac{\sqrt{10x}}{2}$ *is A0*, $\frac{\sqrt{10}}{2}x$ *written as* $\frac{\sqrt{10}x}{2}$ *is A1*
(b)	M1	Uses the correct Pythagoras theorem or correct trigonometry with the given length VA= $\frac{\sqrt{10}}{2}x$ to find the height of the pyramid.
	A1	For $h = \frac{\sqrt{5}}{2}x$ Correct exact answer only scores M1A1. This can be written down as the triangle is isosceles.
(c)	M1	**Ignore missing *x* consistently throughout their solution.** For using any appropriate trigonometry to find the angle *VBA* If attempts cosine rule, look for correct application of the cosine rule for the required angle: $\left(\frac{\sqrt{10}}{2}x\right)^2 = \left(\frac{\sqrt{10}}{2}x\right)^2 + (2x)^2 - 2\times\frac{\sqrt{10}}{2}x\times 2x\cos(VBA)$ oe
	A1	For the correct angle. awrt 50.8 ° (with or without degree sign)
(d)	B1	Identifies the angle required.(can be implied/embedded in their working)
	B1	For the correct lengths AM and MB **or** DM and MC **or** BE and ME Let E to be mid point of AB, correct length BE=*x*, ME=0.5*x* scores B1, this mark can be implied/embedded in their working)
	M1	**Ignore missing *x* consistently throughout their solution.** For using any appropriate trigonometry to find the required angle or half of the required angle.
	A1	For the correct angle. awrt 126.9° (with or without degree sign)
(e)	M1	For using the correct formula for the volume of a pyramid with their height of the pyramid and attempts to find a value of *x*.
	A1	For $x = 3$ with or without unit, ignore incorrect unit

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 18 of 22 · Question 10 part a · 3 marks

2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 19 of 22 · Question 10 part b · 2 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

10(a)	$AC = \sqrt{(2x)^2 + x^2} = (\sqrt{5}x)$ $VA = '\sqrt{5}x' \sin 45^o = \frac{\sqrt{10}}{2}x$ *	B1 M1A1 cso [3]
(b)	Height $= \sqrt{\left(\frac{\sqrt{10}}{2}x\right)^2 - \left(\frac{\sqrt{5}}{2}x\right)^2} = \frac{\sqrt{5}}{2}x$	M1A1 [2]
(c)	$\cos \angle VBA = \frac{1}{\frac{\sqrt{10}}{2}}$ $\Rightarrow \angle VBA = 50.768...^o \approx 50.8^o$ awrt	M1 A1 [2]
(d)	Let M be the point at which the diagonals AC and BD meet. The required angle is either AMB or DMC AM = MB $= \frac{AC}{2} = \frac{\sqrt{5}x}{2}$ or DM=MC=$\frac{AC}{2} = \frac{\sqrt{5}x}{2}$ $\cos \angle AMB = \frac{\left(\frac{\sqrt{5}}{2}\right)^2 + \left(\frac{\sqrt{5}}{2}\right)^2 - 2^2}{2 \times \frac{\sqrt{5}}{2} \times \frac{\sqrt{5}}{2}}$ oe $(\cos \angle AMB = -\frac{3}{5} \Rightarrow \angle AMB = 126.869...^o) \approx 126.9^o$ awrt	B1 B1 M1 A1 [4]
(e)	$9\sqrt{5} = \frac{1}{3} \times 2x \times x \times "\frac{\sqrt{5}}{2}x" \Rightarrow x = \cdots$ $x = 3$	M1 A1 [2]
		Total 13 marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 19 of 22 · Question 10 part b · 2 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For using correct Pythagoras theorem to find length *AC or ½ AC*, explicit substitution must be seen.
	M1	For correct trigonometry using their AC in triangle VAC or half of their AC in triangle VAM or correct Pythagoras theorem (let M be the centre of the base at which the diagonals *AC* and *BD* meet) to find length *VA*. *e.g.* $VA = \sqrt{5}x\cos(45°) = \frac{\sqrt{10}}{2}x$ *or* $VA = \frac{\sqrt{5}}{2}x \div \cos(45°) = \frac{\sqrt{10}}{2}x$ *or* $VA = \frac{\sqrt{5}}{2}x \div \sin(45°) = \frac{\sqrt{10}}{2}x$
	A1 cso	Achieves printed answer with no errors. Condone AV instead of VA Note: $\frac{\sqrt{10}}{2}x$ *written as* $\frac{\sqrt{10x}}{2}$ *is A0*, $\frac{\sqrt{10}}{2}x$ *written as* $\frac{\sqrt{10}x}{2}$ *is A1*
(b)	M1	Uses the correct Pythagoras theorem or correct trigonometry with the given length VA= $\frac{\sqrt{10}}{2}x$ to find the height of the pyramid.
	A1	For $h = \frac{\sqrt{5}}{2}x$ Correct exact answer only scores M1A1. This can be written down as the triangle is isosceles.
(c)	M1	**Ignore missing *x* consistently throughout their solution.** For using any appropriate trigonometry to find the angle *VBA* If attempts cosine rule, look for correct application of the cosine rule for the required angle: $\left(\frac{\sqrt{10}}{2}x\right)^2 = \left(\frac{\sqrt{10}}{2}x\right)^2 + (2x)^2 - 2\times\frac{\sqrt{10}}{2}x\times 2x\cos(VBA)$ oe
	A1	For the correct angle. awrt 50.8 ° (with or without degree sign)
(d)	B1	Identifies the angle required.(can be implied/embedded in their working)
	B1	For the correct lengths AM and MB **or** DM and MC **or** BE and ME Let E to be mid point of AB, correct length BE=*x*, ME=0.5*x* scores B1, this mark can be implied/embedded in their working)
	M1	**Ignore missing *x* consistently throughout their solution.** For using any appropriate trigonometry to find the required angle or half of the required angle.
	A1	For the correct angle. awrt 126.9° (with or without degree sign)
(e)	M1	For using the correct formula for the volume of a pyramid with their height of the pyramid and attempts to find a value of *x*.
	A1	For *x* = 3 with or without unit, ignore incorrect unit

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 19 of 22 · Question 10 part b · 2 marks

2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 20 of 22 · Question 9 part b · 2 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

9(a)	$AC = \sqrt{12^2 + 12^2} = \sqrt{288} = 12\sqrt{2}$	M1A1 [2]
(b)	$x = \dfrac{\dfrac{12\sqrt{2}}{2}}{\cos 30^\circ} = \sqrt{96} = \left(4\sqrt{6}\right)^*$	M1A1 cso [2]
(c)	$\cos \angle AOB = \dfrac{96 + 96 - 12^2}{2 \times \sqrt{96} \times \sqrt{96}} \Rightarrow \angle AOB = 75.522...^\circ$ Area $\Delta OAB = \dfrac{1}{2} \times \sqrt{96} \times \sqrt{96} \times \sin' 75.522^\circ ' = \left(46.4758...\right)$ $4 \times 46.4758 + 12^2 = 329.90.... \Rightarrow$ Area = 330 [m^2]	M1A1 M1 M1A1 [5]
(d)	$\left[\angle OBC = \dfrac{180 - 75.522}{2} = 52.239^\circ\right]$ Let Y on OB be the foot of the perpendicular from A to OB $AY = 12 \sin 52.239^\circ = \left(9.4868...\right)$ $\cos \angle AYC = \dfrac{'9.4868'^2 + '9.4868'^2 - 288}{2 \times '9.4868' \times '9.4868'} \Rightarrow \angle AYC = 126.87... \approx 127^{[o]}$	M1 A1 M1A1 [4]
		Total 13 marks

Question	Notes	Marks
9(a)	For using Pythagoras theorem on triangle ABC or triangle ADC $AC = \sqrt{12^2 + 12^2} = ...$	M1
	For the correct value of AC $AC = \sqrt{288} = 12\sqrt{2}$	A1 [2]
(b)	Let the intersection of AC and BD be X Uses any appropriate trigonometry on triangle OAX For example; $x = \dfrac{\dfrac{12\sqrt{2}}{2}}{\cos 30^\circ} = ...$ **OR** $\dfrac{\sin 120^\circ}{12\sqrt{2}} = \dfrac{\sin 30^\circ}{OA} \Rightarrow OA = ...$ OR $\left(12\sqrt{2}\right)^2 = x^2 + x^2 - 2 \times x \times x \cos 120 \Rightarrow x = ...$	M1
	For the correct value of x	A1 cso

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 20 of 22 · Question 9 part b · 2 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

	$x = \sqrt{96} = \left(4\sqrt{6}\right)$ *	[2]
(c)	For using trigonometry to find angle $\angle AOB$ $\cos \angle AOB = \frac{96+96-12^2}{2\times\sqrt{96}\times\sqrt{96}} = ...$	M1
	$\angle AOB = 75.522...^o$	A1
	Area of triangle OAB Area $= \frac{1}{2}\times\sqrt{96}\times\sqrt{96}\times\sin'75.522^o{}' = \left(46.4758...\right)$	M1
	Total area of the pyramid $= 4\times46.4758+12^2 = \left(329.90....\right)$	M1
	For the correct final area = awrt 330 (cm^2)	A1 [5]
	ALT	
	Height of the triangle of one of the triangular faces: $h = \sqrt{\left(4\sqrt{6}\right)^2 - 6^2} = ...$	M1
	$h = 2\sqrt{15}$	A1
	Area of triangle $OAB = \frac{1}{2}\times2\sqrt{15}\times12 = 12\sqrt{15} = \left(46.4758...\right)$	M1
	Total area of the pyramid $= 4\times12\sqrt{15}+144 = \left(329.903...\right)$	M1
	For the correct final area = awrt 330 (cm^2) Allow an exact answer of $48\sqrt{15}+144$ $\left(cm\right)^2$ oe	A1
(d)	$\cos \angle AOB = \frac{96+96-12^2}{2\times\sqrt{96}\times\sqrt{96}} = 75.522^o$ or $\angle OBC = \frac{180^o - 75.522^o}{2} = 52.239^o$	
	Let Y on OB be the foot of the perpendicular from A to OB Length $AY = 12\sin 52.239^o = \left(9.4868...\right)$ OR Length $AY = 4\sqrt{6}\sin 75.522^o = \left(9.4868...\right)$	M1A1 [M1A1]
	For the appropriate trigonometry on triangle AYC to find angle AYC $\cos \angle AYC = \frac{'9.4868'^2+'9.4868'^2-288}{2\times'9.4868'\times'9.4868'} \Rightarrow \left(\angle AYC = 126.87\right)$	M1
	Angle between plane AOB and plane OBC =awrt 127[°]	A1 [4]
		Total 13 marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 20 of 22 · Question 9 part b · 2 marks

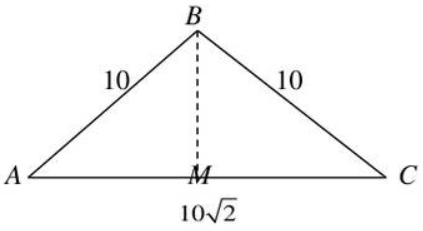
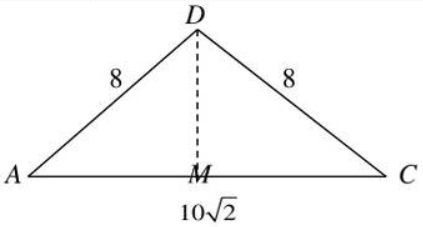
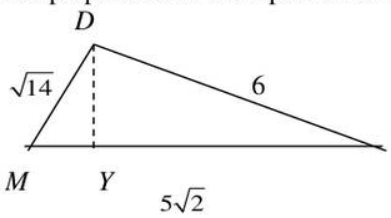
2023 JAN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 21 of 22 · Question 9 part d · 2 marks

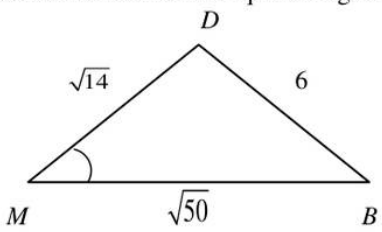
2021 NOV · Paper 02 · extract 1 of 3 · find part d in this mark scheme extract

9 (a)	$AC = \sqrt{10^2 + 10^2} = \sqrt{200} = 10\sqrt{2}$	M1A1 [2]
(b)	 $BM = \sqrt{10^2 - \left(5\sqrt{2}\right)^2} = 5\sqrt{2}$	M1A1 [2]
(c)	 $DM = \sqrt{8^2 - \left(5\sqrt{2}\right)^2} = \sqrt{14}$ $\angle DMB = \cos^{-1}\left(\frac{14 + 50 - 36}{2 \times \sqrt{14} \times 5\sqrt{2}}\right) = 58.0519...^\circ \approx 58.1^\circ$	M1A1 M1A1 [4]
(d)	Let perpendicular from point D to BM be the point Y  Vertical height of shape $ABCD$ In triangle DMY (above) Height $= DY = \sqrt{14}\sin 58.0519^\circ = 3.1749... \approx 3.17$ (cm)	M1A1 [2]
		Total 10 marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 21 of 22 · Question 9 part d · 2 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part d in this mark scheme extract

Part	Mark	Notes
(a)	M1	For using Pythagoras theorem correctly on triangle ABC to find AC
	A1	For $AC = 10\sqrt{2}$
(b)	M1	For using Pythagoras theorem or any appropriate trigonometry correctly on triangle ABC to find BM
	A1	For the correct length $BM = 5\sqrt{2}$ **NB:** Allow for the answer just seen without any working as the triangle ABC is isosceles.
(c)	M1	For using Pythagoras theorem to find the length DM
	A1	For $DM = \sqrt{14}$ or accept awrt 3.74
	M1	For using cosine rule to find the required angle DMB 
	A1	For awrt 58.1°
(d)	M1	For using any appropriate trigonometry to find the required length. Ft their angle DMB
	A1	For awrt 3.17 (cm) Accept 3.18 (cm)

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 21 of 22 · Question 9 part d · 2 marks

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Question number	Scheme	Marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 22 of 22 · Question 9 part c · 3 marks

2024 JUN · Paper 02 · extract 1 of 4 · find part c in this mark scheme extract

9(a)	$\sin 75^\circ = \sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \sin 30^\circ \cos 45^\circ$ $= \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{\sqrt{2}}{2} = \frac{\sqrt{6}+\sqrt{2}}{4}$ *	M1 M1A1 cso
	ALT 1 $\sin 75^\circ = \sin(120^\circ - 45^\circ) = \sin 120^\circ \cos 45^\circ - \sin 45^\circ \cos 120^\circ$ $= \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \times \left(-\frac{1}{2}\right) = \frac{\sqrt{6}+\sqrt{2}}{4}$ *	[3] [M1 M1A1]
	ALT 2 $\sin 75^\circ = \sin(60^\circ + 15^\circ) = \sin 60^\circ \cos 15^\circ + \sin 15^\circ \cos 60^\circ$ $= \frac{\sqrt{3}}{2} \times \left(\frac{\sqrt{6}+\sqrt{2}}{4}\right) + \left(\frac{\sqrt{6}-\sqrt{2}}{4}\right) \times \frac{1}{2}$ $= \frac{3\sqrt{2}+\sqrt{6}}{8} + \frac{\sqrt{6}-\sqrt{2}}{8} = \frac{\sqrt{6}+\sqrt{2}}{4}$	[M1 M1 A1]
(b)	$\left[\angle AFC = 180^\circ - 60^\circ - 45^\circ = 75^\circ\right]$ $\frac{24}{\sin 75^\circ} = \frac{BF}{\sin 60^\circ}$ $\Rightarrow \frac{24}{\sin 75^\circ} = \frac{BF}{\sin 60^\circ} \Rightarrow \frac{24}{\frac{\sqrt{6}+\sqrt{2}}{4}} = \frac{BF}{\frac{\sqrt{3}}{2}}, \Rightarrow BF = \frac{48\sqrt{3}}{\sqrt{6}+\sqrt{2}}$ $BF = \frac{48\sqrt{3}}{(\sqrt{6}+\sqrt{2})} \times \frac{(\sqrt{6}-\sqrt{2})}{(\sqrt{6}-\sqrt{2})} = \frac{48\sqrt{3}\sqrt{6}-48\sqrt{6}}{6-2} = 12(3\sqrt{2}-\sqrt{6})$ *	M1,M1A1 M1A1 cso [5]
(c)	Let the vertical point from F to the line AB be X and the corresponding point on CD be Y. The required angle is EXY $FX = 12(3\sqrt{2}-\sqrt{6})\sin 45^\circ = 36 - 12\sqrt{3} = (15.2153...)$ OR Area $ABC = \frac{1}{2} \times 24 \times FX = \frac{1}{2} \times 24 \times 12(3\sqrt{2}-\sqrt{6})\sin 45$ $\Rightarrow FX = 12(3\sqrt{2}-\sqrt{6})\sin 45^\circ = 36 - 12\sqrt{3} = (15.2153...)$ $EF = \frac{36-12\sqrt{3}}{\tan 65^\circ} = 7.09505.... \approx 7.1$ (cm)	M1 M1A1 [3]
(d)	Identifies the required angle FCX $FC = \sqrt{\left[12(3\sqrt{2}-\sqrt{6})\right]^2 + '7.09505'^2} = 22.65735.....$ $\angle FCX = \sin^{-1}\left(\frac{15.2153}{'22.6573'}\right) = 42.186...^\circ \approx 42.2^\circ$	B1 M1 M1A1 [4]
		Total 15 marks

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 22 of 22 · Question 9 part c · 3 marks

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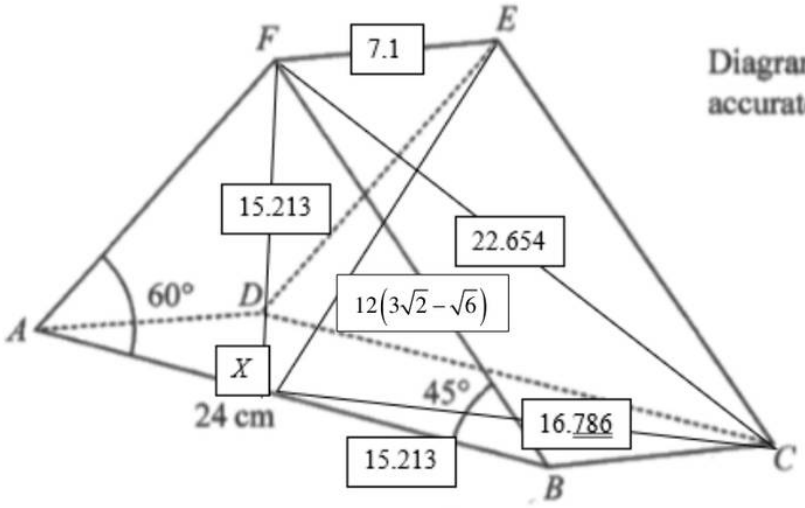
Part	Mark	Notes
(a)	M1	For the correct use of the addition formula for sine with $\sin(45^\circ+30^\circ)$ Accept also $\sin(120^\circ-45^\circ)$ and $\sin(60^\circ+15^\circ)$ **but not** $\sin(90^\circ-15^\circ)$ or $\sin(180^\circ-105^\circ)$
	M1	For applying the exact values of these sines and cosines and for evaluating the expression.. This step must be explicitly seen.
	A1 cso	For the correct final answer with no errors seen
(b)	M1	For the correct use of sine rule using the exact value for sin 75° and sin 60°
	M1	For rearranging to obtain $BF=\frac{k\sqrt{3}}{\sqrt{6}+\sqrt{2}}$ or $\frac{l\sqrt{3}}{2\sqrt{6}+2\sqrt{2}}$ Where k and l are integers
	A1	For the correct **exact** BF which need not be simplified. $BF=\frac{48\sqrt{3}}{\sqrt{6}+\sqrt{2}}$ or $\frac{96\sqrt{3}}{2\sqrt{6}+2\sqrt{2}}$ or even $\frac{12\sqrt{3}\times4}{\sqrt{6}+\sqrt{2}}$ but not for example, $\frac{\frac{12\sqrt{3}}{\sqrt{6}+\sqrt{2}}}{4}$ unless recovery is seen later
	M1	For multiplying numerator **and** denominator by their conjugate which must of the form e.g $\sqrt{A}\pm\sqrt{B}$ **AND** simplifying to the required form. We need to see at least one intermediate line of working before the final answer here.
	A1 cso	The answer as shown with no errors or omissions seen. This is a given result.
(c)	M1	For finding the length FX using any appropriate trigonometry.
	M1	For using any appropriate trigonometry to find length EF [or XY]
	A1	For awrt 7.1 (cm)
(d)	B1	For identifying the correct angle required.
	M1	For using Pythagoras theorem to find FC $FC=\sqrt{21.51781^2+7.09505^2}=22.65735.....$ Or For using Pythagoras theorem to find CX $FC=\sqrt{15.213^2+7.095^2}=16.786...$ [Note: $BX=15.213...$ and $CX=16.786\ldots$]
	M1	For using any appropriate trigonometry to find angle FCX. For example; $\angle FCX=\tan^{-1}\left(\frac{15.2153}{16.786}\right)=42.186...^\circ\approx42.2^\circ$
	A1	For awrt 42.2°

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 22 of 22 · Question 9 part c · 3 marks

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USEFUL SKETCH



NB:

$AC = 25.03$ cm

$AF = 17.57$ cm

$CF = 22.65$ cm

Angle $FCA = 42.9°$

Type MS · Lengths in a 3D solid using right-angled triangles

Answer 22 of 22 · Question 9 part c · 3 marks

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Question	Scheme	Marks